



Basic Mechanical Engineering

~ ME2001 ~

Semester: Spring 2023
Department of Electrical Engineering
National University of Computer and Emerging Sciences
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Introduction

- ❑ INSTRUCTOR
- ❑ BASIC MECHANICAL ENGINEERING
- ❑ INTRODUCTION





Instructor

TEACHING METHODOLOGY





Instructor

SLIDE THEME

Questions, Disadvantages & Warnings

Answers & Advantages

Definitions & Facts

[1] - References



Basic Mechanical Engineering

IDEA

What does “Engineering” Mean?

Application of scientific, economic, social, and practical knowledge in order to invent, design, build, maintain, research, and improve structures, machines, devices, systems, materials and processes.

What is Mechanical Engineering?

Discipline that applies the principles of engineering, physics and materials science for the design, analysis, manufacturing, and maintenance of mechanical systems

What is meant by Mechanical Systems?

A mechanical system manages power to accomplish a task that involves forces and movement.



Basic Mechanical Engineering

ENGINEER

What does an Engineer do?

Applies scientific knowledge and ingenuity to
develop solutions for technical, societal and
commercial problems



Cleverness

What are you doing here?

Learning scientific knowledge and
its applications

YOU ALL ARE PROBLEM SOLVERS



wikiHow



Basic Mechanical Engineering

APPLICATIONS



The Astounding Athletic Power of Quad copters

[1] - Raffaello D'Andrea: The astounding athletic power of quadcopters, TEDGlobal (JUN 2013), URL :<http://on.ted.com/Quadcopter>



Basic Mechanical Engineering

APPLICATIONS



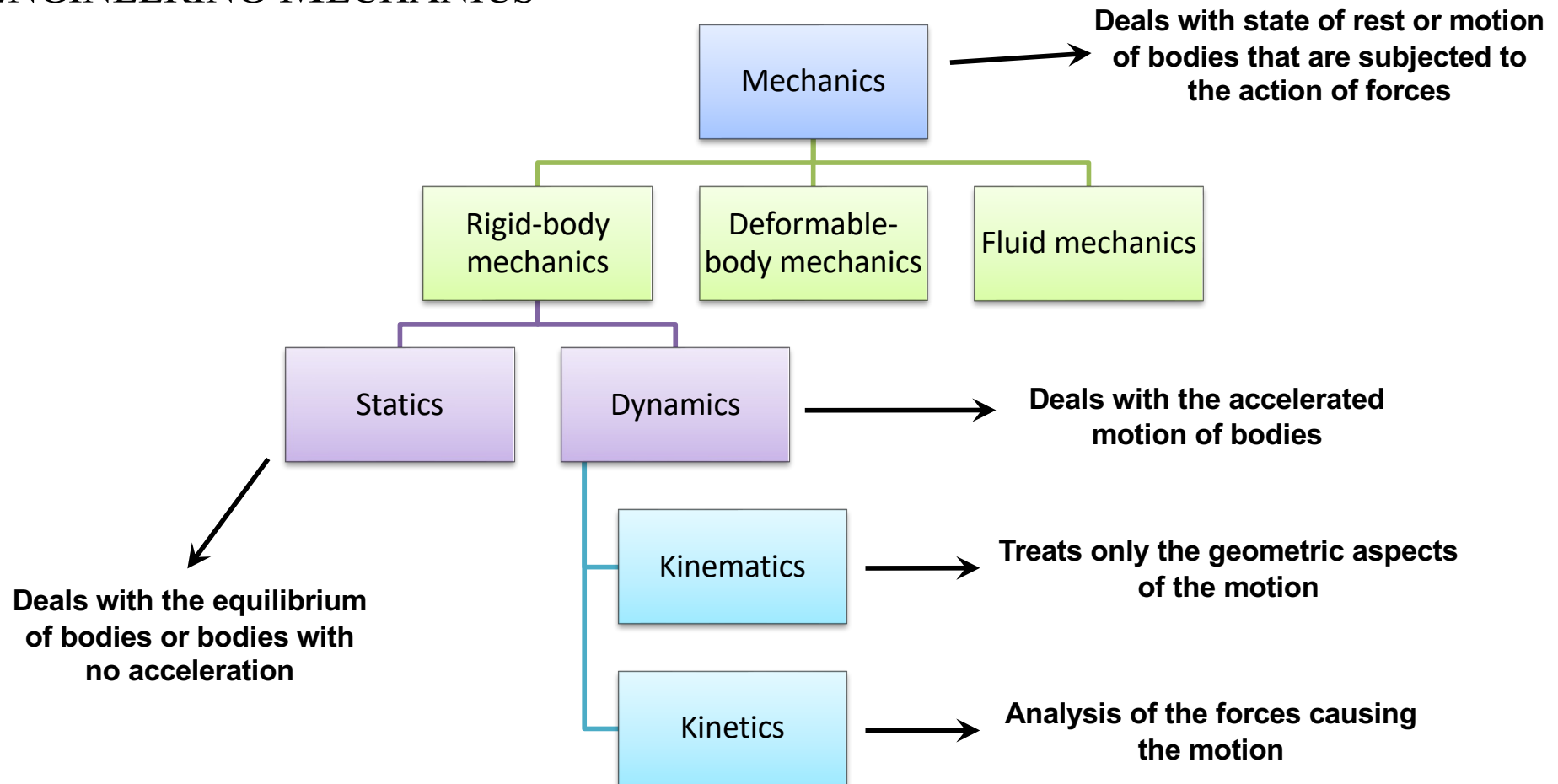
Engineering Mechanics is the study of forces and movements [1]

[1] – Youtube – January 1, 2015 (www.youtube.com/watch?v=RFFidyoyDmk)



Basic Mechanical Engineering

ENGINEERING MECHANICS



Essential for the design and analysis of many types of structural members, mechanical components, or electrical devices encountered in engineering



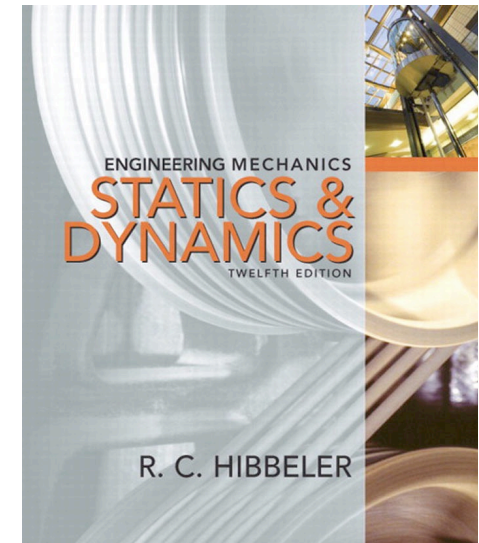
Basic Mechanical Engineering

OBJECTIVE

To build the fundamental concepts of engineering mechanics and to clearly present their application in the analysis of planner motion of mechanical components and structures.

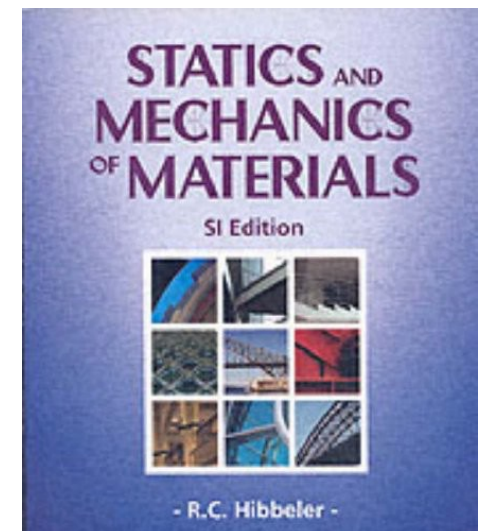
COURSE BOOK

- ✓ *Engineering Mechanics: Statics and Dynamic*, by R.C Hibbeler, Pearson Prentice Hall, 12th Edition, 2010.



REFERENCE BOOKS

- ✓ *Statics and Mechanics of Materials*, by R.C Hibbeler, Pearson Prentice Hall, SI Edition, 2004.
- ✓ *Engineering Mechanics: Statics and Dynamics*, by J. L. Meriam, and L. G. Kraige, John & Wiley Sons , 7th Edition,, 2012





Basic Mechanical Engineering

COURSE OUTLINE

Week Number	Book Topics Numbers	Topic Names
1	1.1 to 1.3	Introduction to Vector Mechanics
2	2.1 to 2.7	Force Systems
3	3.1 to 3.2	Free Body Diagram + Quiz I
4	3.5 to 3.10	Equilibrium of a Rigid Body
5	5.1 to 5.2	Friction
6	Midterm Examination I	

Appropriate changes would be made if required.



Basic Mechanical Engineering

COURSE OUTLINE

Week Number	Book Topics Numbers	Topic Names
7	16.1 to 16.4	Planar Kinematics: Absolute Motion Analysis
8	16.5, 16.7	Planar Kinematics : Relative Motion Analysis
9	17	Planar Kinetics: Force and Acceleration
10	17	Planar Kinetics: Force and Acceleration + Quiz II
11	18	Planar Kinetics: Work and Energy
12	Midterm Examination II	

Appropriate changes would be made if required.



Basic Mechanical Engineering

COURSE OUTLINE

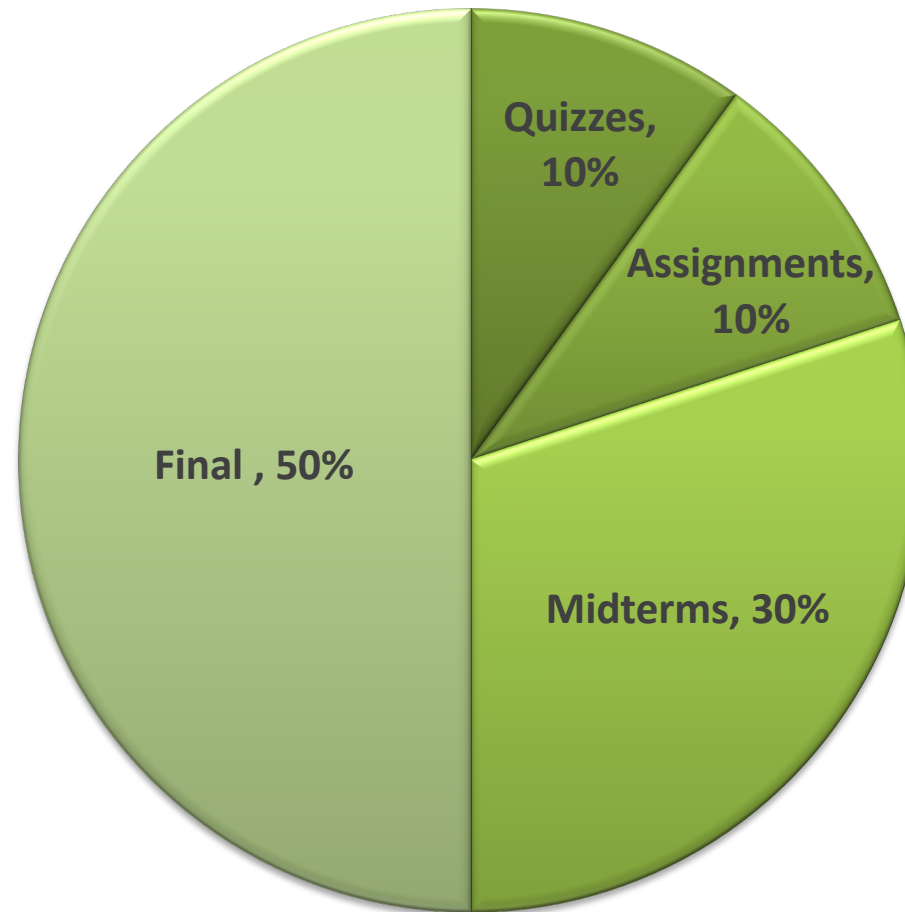
Week Number	Book Topics Numbers	Topic Names
13	16.1 to 16.4	Mechanics of Materials: Stress and Strain
14	16.5, 16.7	Mechanics of Materials: Torsion & Bending
15	17	Mechanical Properties of Material + Quiz III
16	18	Structure Analysis: Frames and Machines
		Final Examination

Appropriate changes would be made if required.



Basic Mechanical Engineering

GRADING POLICY



Bonus, 5% Absolute (Max)



Basic Mechanical Engineering

A+ OR F

How to succeed in this course?

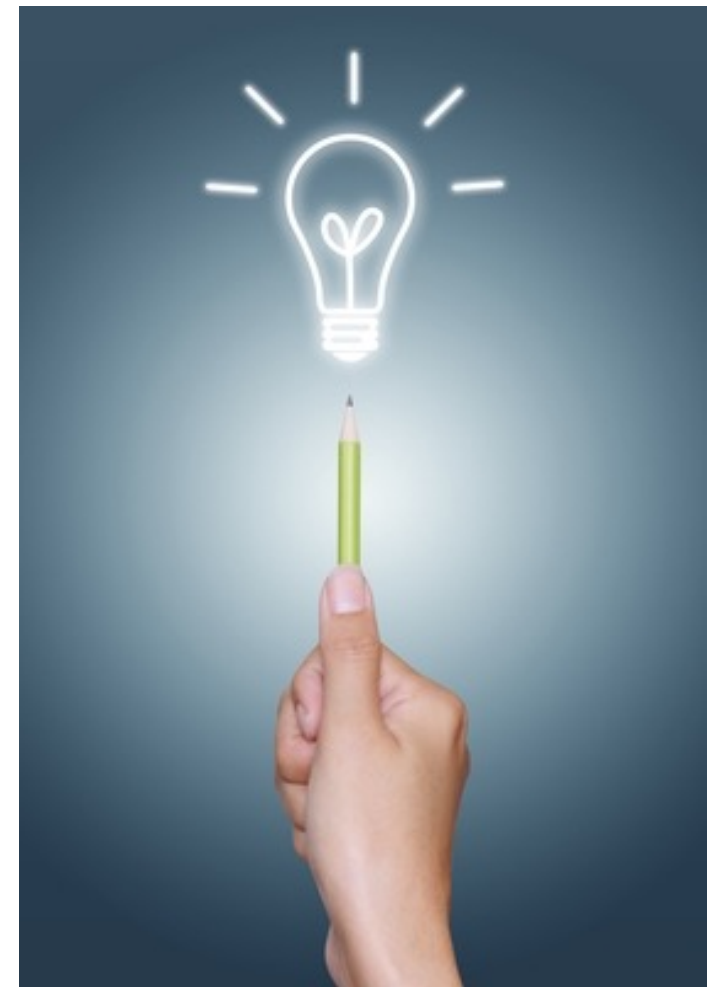
- ✓ Arrive on time
- ✓ Participate in all classes
- ✓ Pay attention, take notes and ask questions if needed
 - ✓ Identify and understand key issues
- ✓ Review material (notes and textbook) before and after class
 - ✓ Avoid Plagiarism
- ✓ Practice, Practice and Practice



Take-Home Message

THOUGHT

You are a Problem Solver



Fundamental Concepts

- ❑ BASIC QUANTITIES
- ❑ COORDINATE SYSTEMS





Basic Quantities

MEASURING OBJECTS

How do we measure the size of a physical object?

Lengths & Angles

What are the units?

Meter - Radians

How long is 1 Meter?

One ten-millionth of the distance from the Earth's equator to the North Pole

Length of the path travelled by light in vacuum during a time interval of $1/299,792,458$ of a second





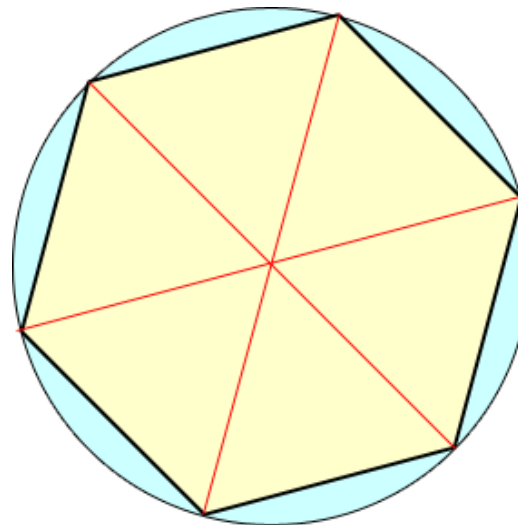
Basic Quantities

MEASURING OBJECTS

Why a circle has 360 degrees?

Why not 248 or 623 degrees?

Babylonians subdivided the circle using the angle of an equilateral triangle as the basic unit and further subdivided the latter into 60 parts



Which one is greater degree or radian?

1 Radian = 57.3 Degrees

Radian > Degrees



Basic Quantities

MEASURING OBJECTS

How much is one Radian?

Numerically equal to the length of a corresponding arc of a unit circle

$$\text{Length of arc } (l) = \text{Radius } (r) \times \text{Angle } (\theta)$$





Basic Quantities

RADIUS OF EARTH

What is the mean radius of earth?

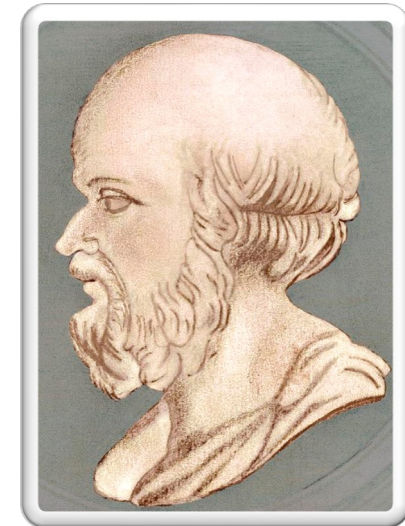
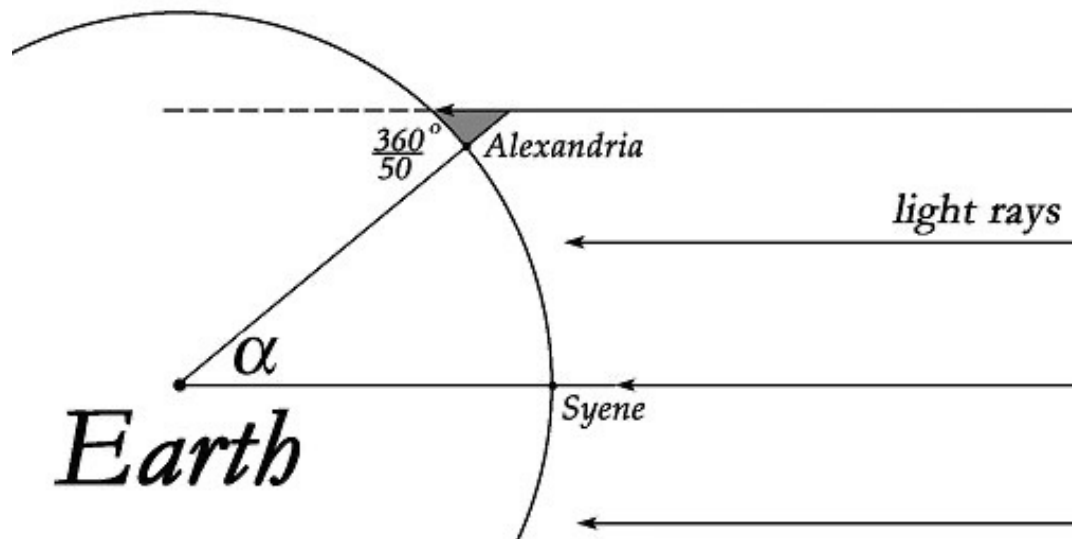
6371 Km

Who calculated it?

Eratosthenes

How?

$$l = r \theta$$



Eratosthenes
(276BC - 194BC)
Greek mathematician,
geographer, poet,
astronomer





Basic Quantities

MEASUREMENT QUANTITY

How do we measure that amount of matter in an object?

Mass

Units?

Kilogram (Kg)

How is it defined?

How much is 1Kg?

1 gram is the mass of 1 cubic centimeter of water

How is it measured?

Inertial Balance

What is the mass of earth?

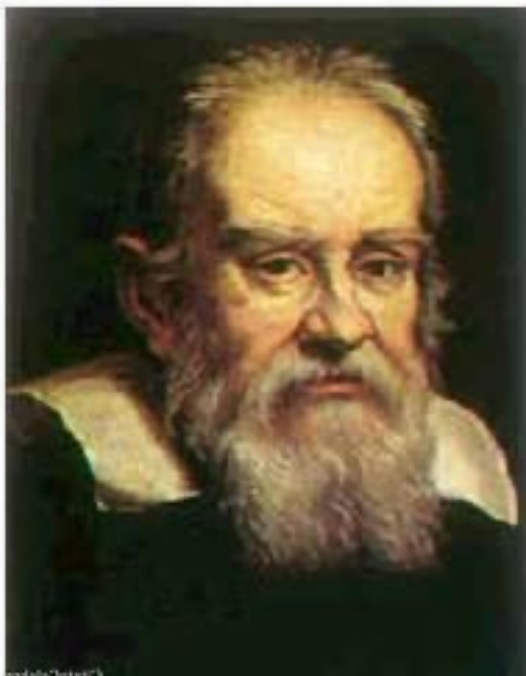
$5.972 \times 10^{24} \text{ Kg}$



Basic Quantities

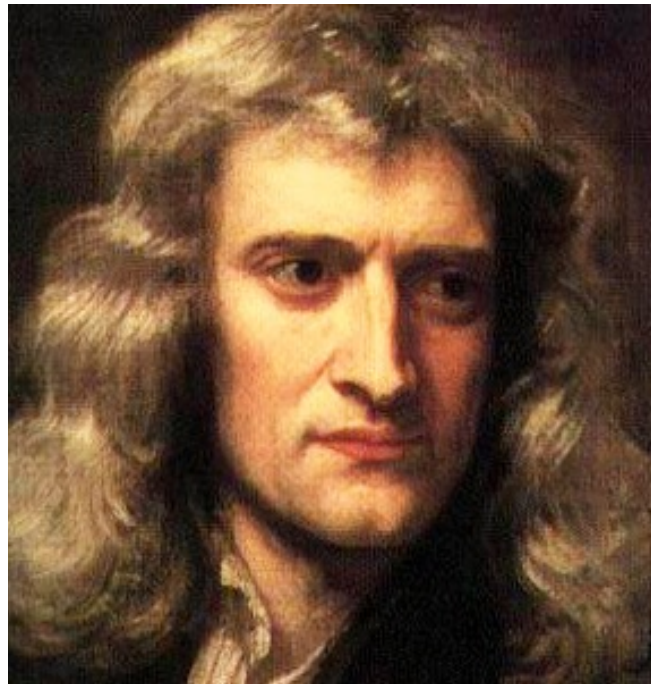
MASS OF EARTH

How was mass of the Earth measured?



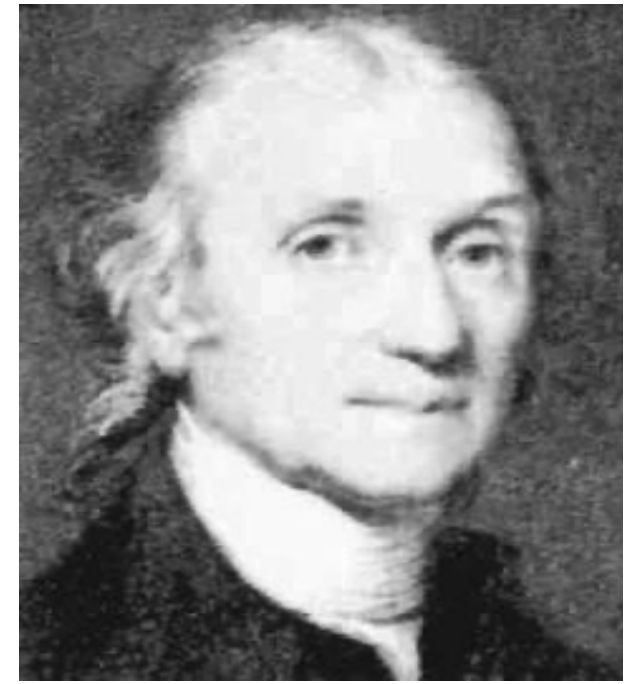
Galileo Galilei
(1564-1642)

$$g = 9.81 \text{ m/s}^2$$



Isaac Newton
(1642-1726)

$$F = \frac{Gm_1m_2}{r^2}$$



Henry Cavendish
(1731-1810)

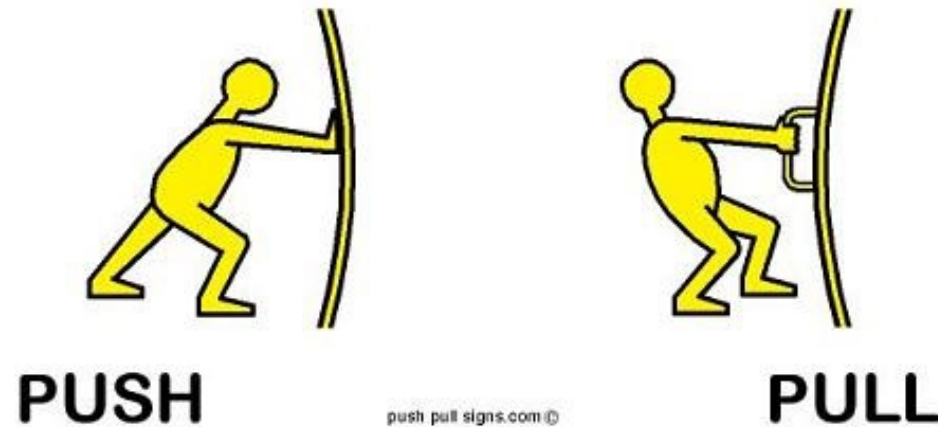
$$G = 66.73 \times 10^{-12} \frac{\text{m}^3}{\text{Kg.s}^2}$$



Basic Quantities

MEASURING PUSH/PULL

Can we measure “Push” or “Pull”?



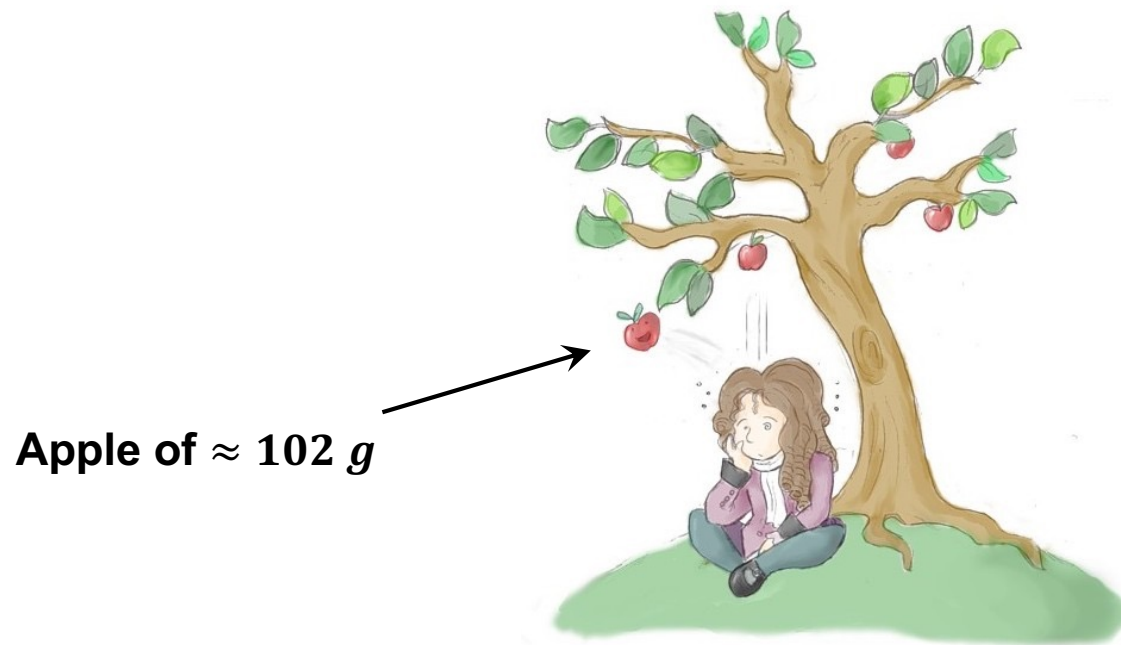
Yes! Force in Newton



Basic Quantities

MEASURING PUSH/PULL

How much is the push/pull of one Newton force?



What is the pulling effect of Earth on an object called?

Weight



Basic Quantities

MEASURING PUSH/PULL

Is weight of an object constant?

No!

Why?

$$F = \frac{Gm_1m_2}{r^2}$$

G = Universal Constant of Gravitation = 66.73×10^{-12}



Coordinate Systems

INTRODUCTION

What do we need to specify the position of an object in 3D space ?

**A reference
A convention for representation**

What do we get by combining the above-mentioned entities?

A coordinate system

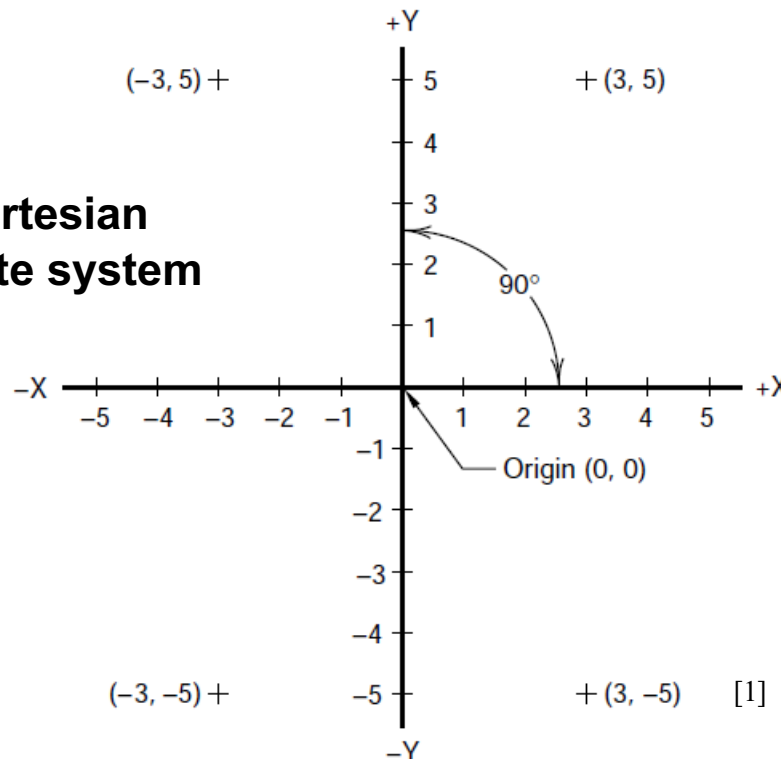


Coordinate Systems

HISTORY

for every point in space, a set of real numbers can be assigned, and for each set of real numbers, there is a unique point in space

2-D Cartesian coordinate system



Renè Descartes (1596–1650)

French Mathematician
“Cartesius”

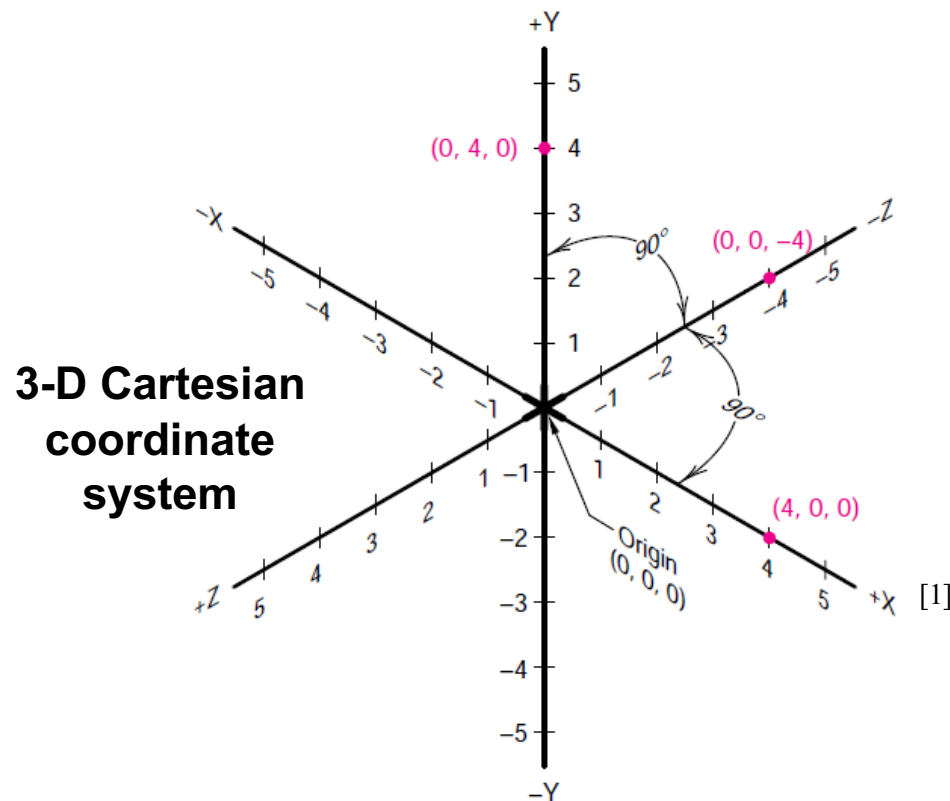
[1] - Gary Bertoline, Eric Wiebe et al, "Technical Graphics Communication", McGraw-Hill Science/Engineering/Math, 4 edition, January 31, 2008, p 307.



Coordinate Systems

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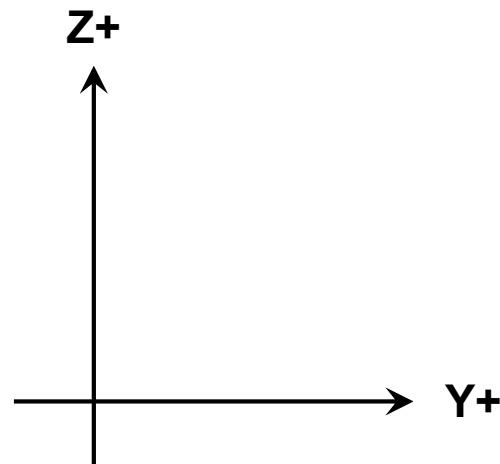
French Mathematician
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[1] - Gary Bertoline, Eric Wiebe et al, "Technical Graphics Communication", McGraw-Hill Science/Engineering/Math, 4 edition, January 31, 2008, p 307.



Coordinate Systems

CARTESIAN COORDINATE SYSTEM

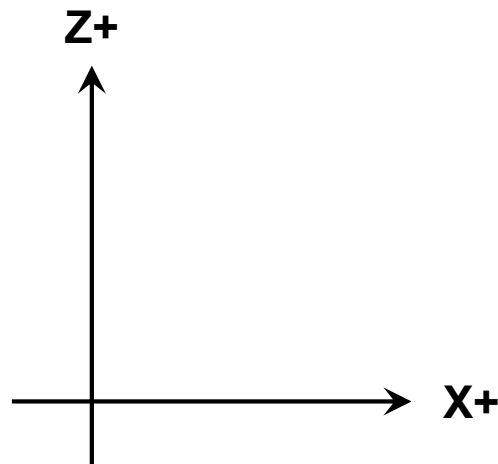


**In which direction is X+ axis?
In the plane or out of the plane?**
Out of the plane



Coordinate Systems

CARTESIAN COORDINATE SYSTEM



**In which direction is Y+ axis?
In the plane or out of the plane?**

In the plane

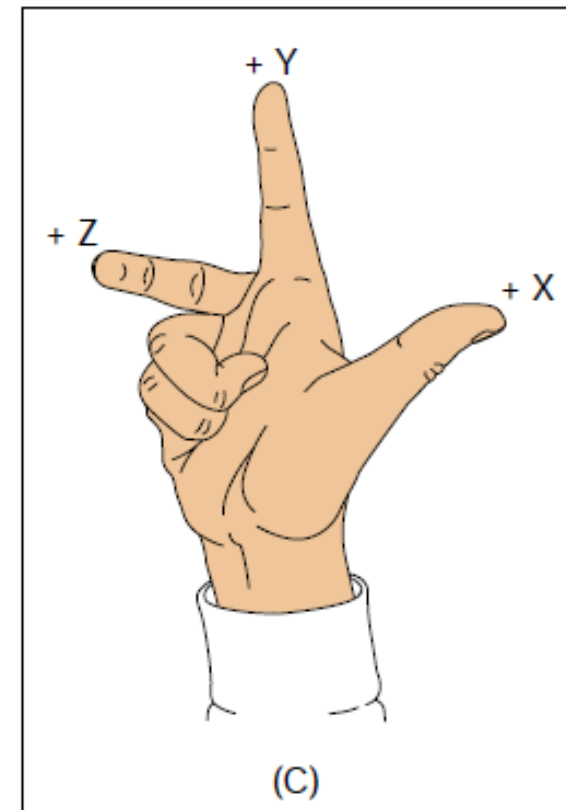
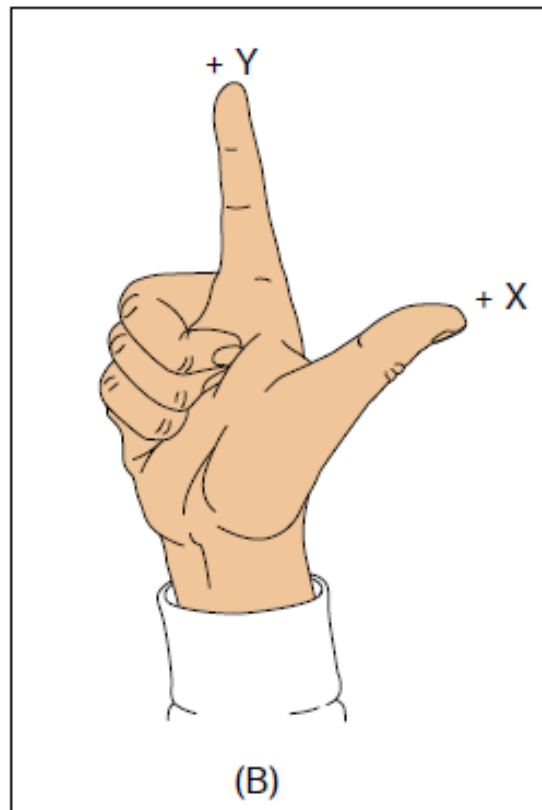
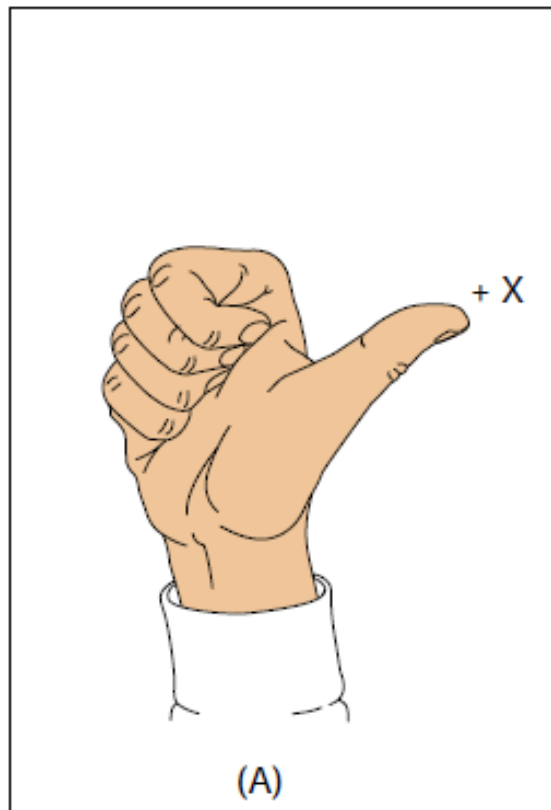
**How do you know for sure?
What is the rule that you are applying?**

Right Hand Rule



Coordinate Systems

RIGHT HAND RULE



[1]

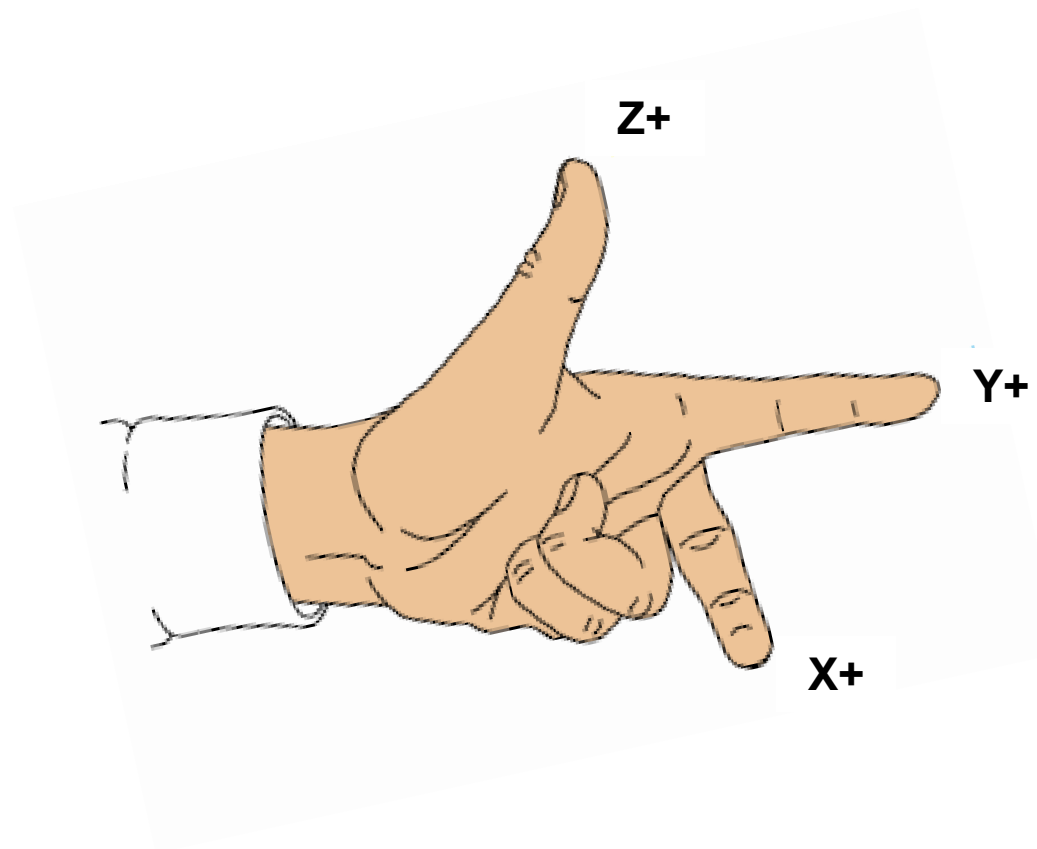
Right-Hand Rule

[1] - Gary Bertoline, Eric Wiebe et al, "Technical Graphics Communication", McGraw-Hill Science/Engineering/Math, 4 edition, January 31, 2008, p 310.



Coordinate Systems

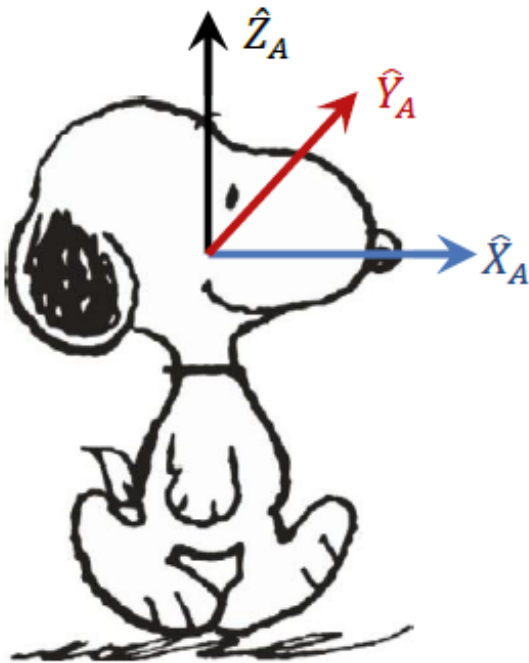
RIGHT HAND RULE



Left Hand Rule (Not Documented)

Coordinate Systems

ROTATION ABOUT AXIS

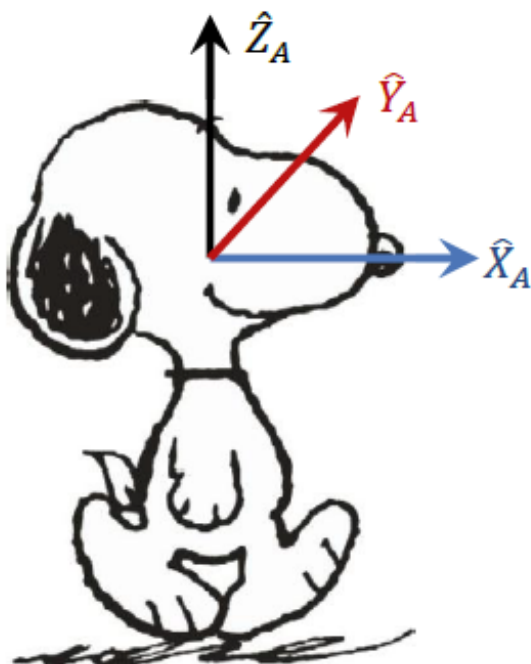


If Snoopy is rotated 90° around Z+ axis which one of the two diagrams show its possible orientations



Coordinate Systems

ROTATION ABOUT AXIS



If Snoopy is rotated 90° around $Y+$ axis which one of the two diagrams show its possible orientations



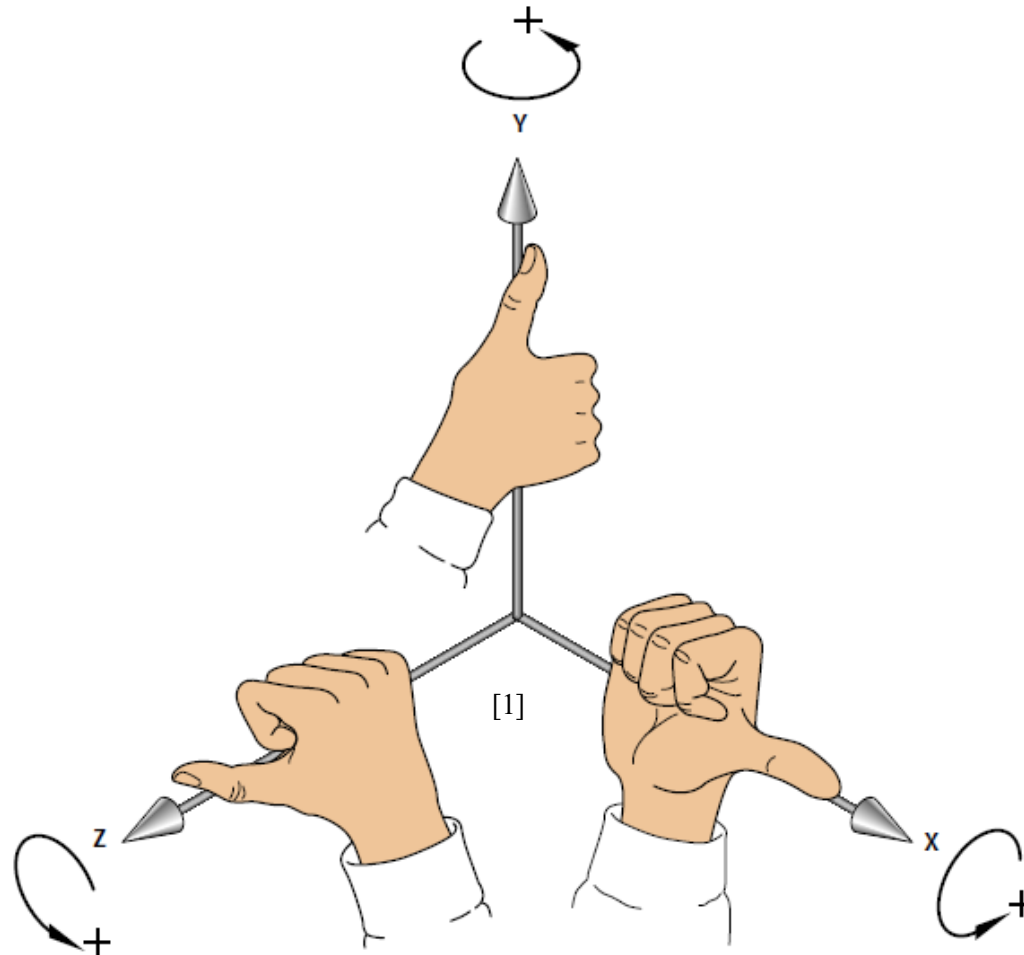
How do you know for sure?
What is the rule that you are applying?

Right Hand Rule



Coordinate Systems

ROTATION ABOUT AXIS – RIGHT HAND RULE



Positive directions of rotation on each axis.

[1] - Gary Bertoline, Eric Wiebe et al, "Technical Graphics Communication", McGraw-Hill Science/Engineering/Math, 4 edition, January 31, 2008, p 310.

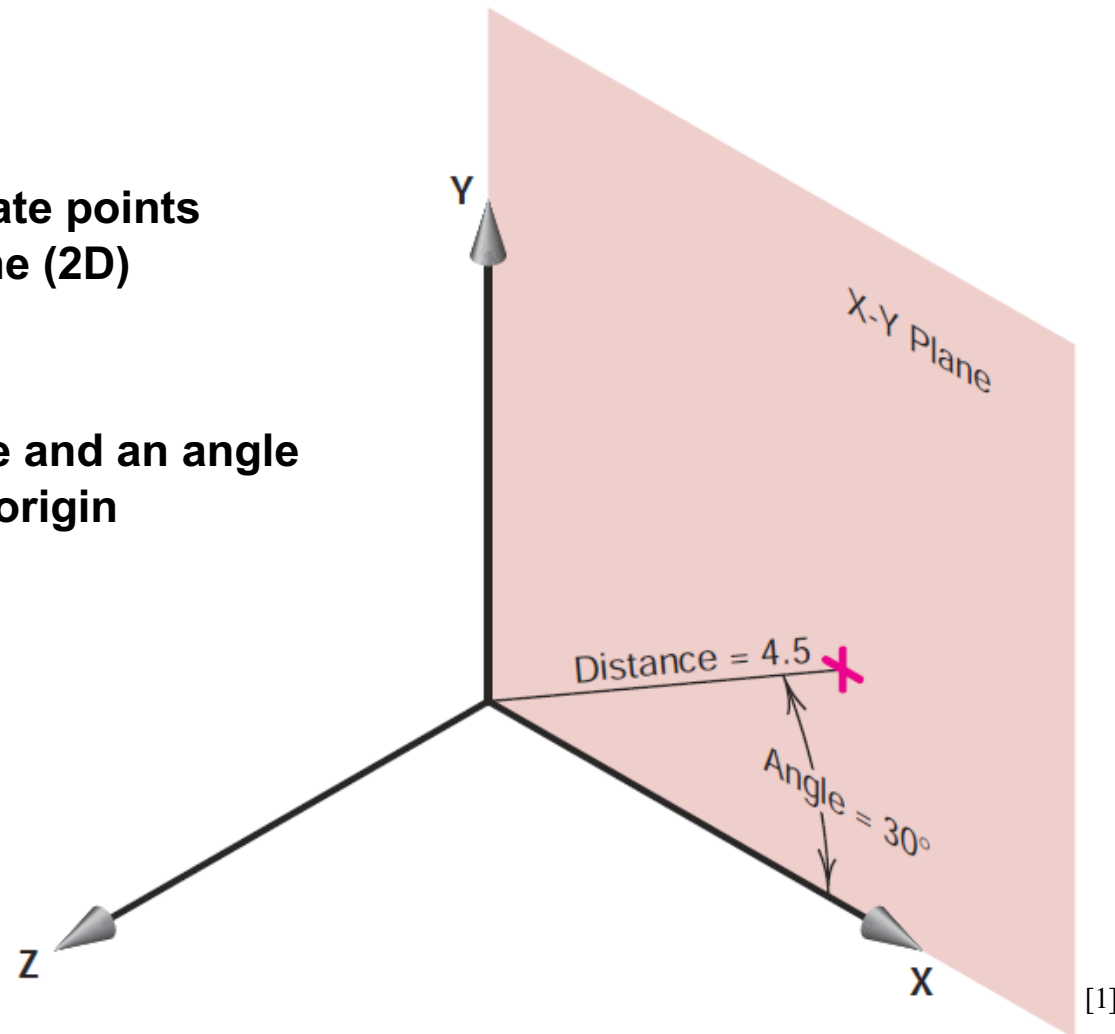


Coordinate Systems

POLAR COORDINATES

**Is used to locate points
in any plane (2D)**

**Specify a distance and an angle
from the origin**



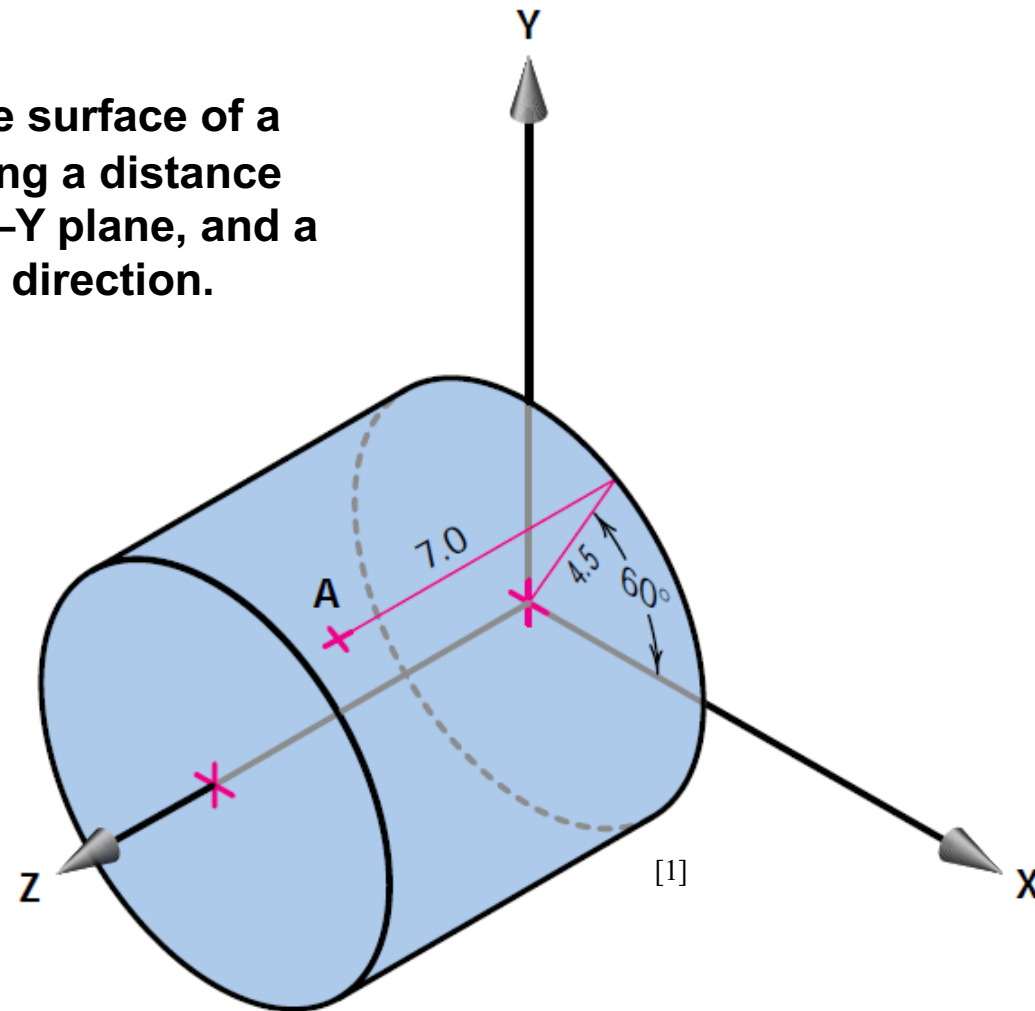
[1] - Gary Bertoline, Eric Wiebe et al, "Technical Graphics Communication", McGraw-Hill Science/Engineering/Math, 4 edition, January 31, 2008, p 311.



Coordinate Systems

CYLINDRICAL COORDINATES

locate a point on the surface of a cylinder by specifying a distance and an angle in the X–Y plane, and a distance in the Z direction.



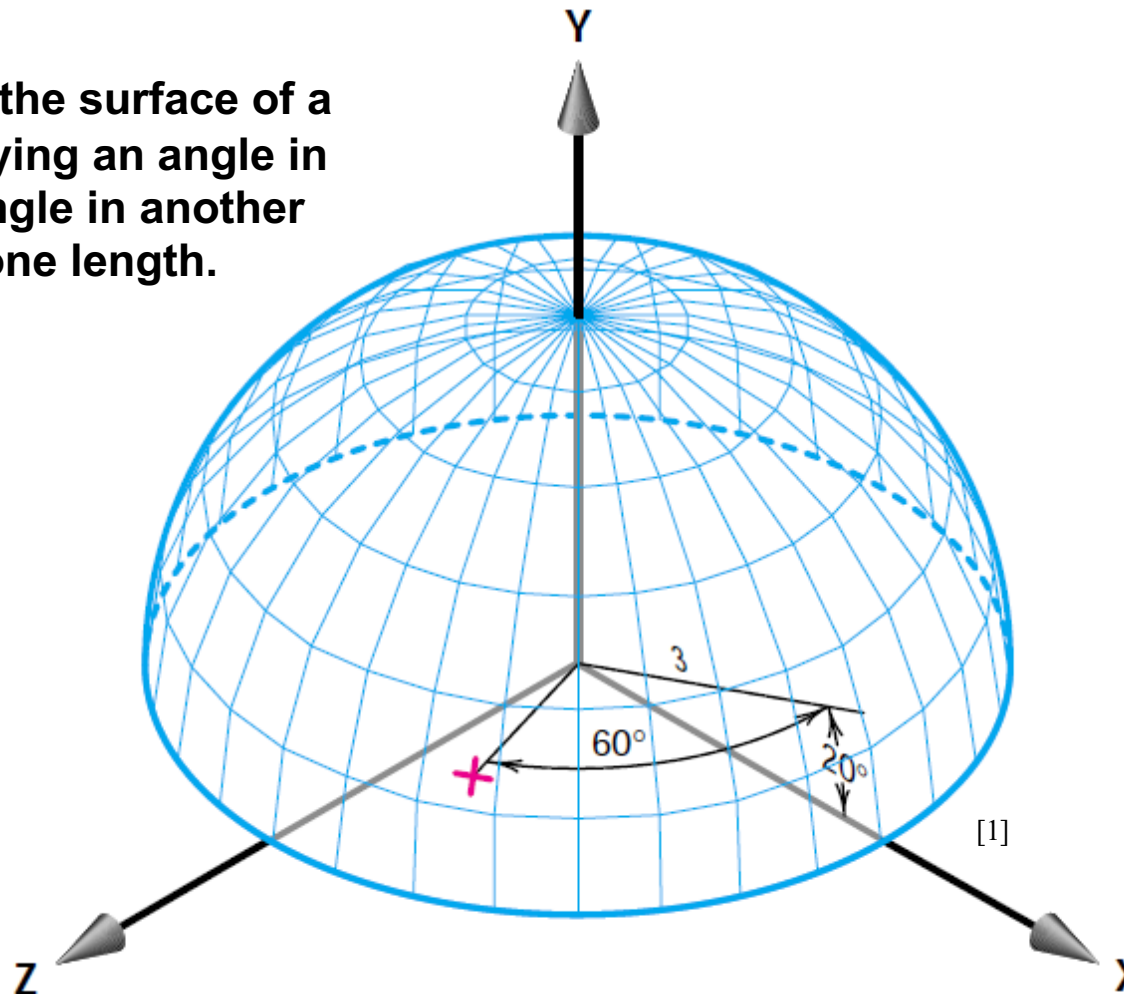
[1] - Gary Bertoline, Eric Wiebe et al, "Technical Graphics Communication", McGraw-Hill Science/Engineering/Math, 4 edition, January 31, 2008, p 311.



Coordinate Systems

SPHERICAL COORDINATES

locate a point on the surface of a sphere by specifying an angle in one plane, an angle in another plane, and one length.

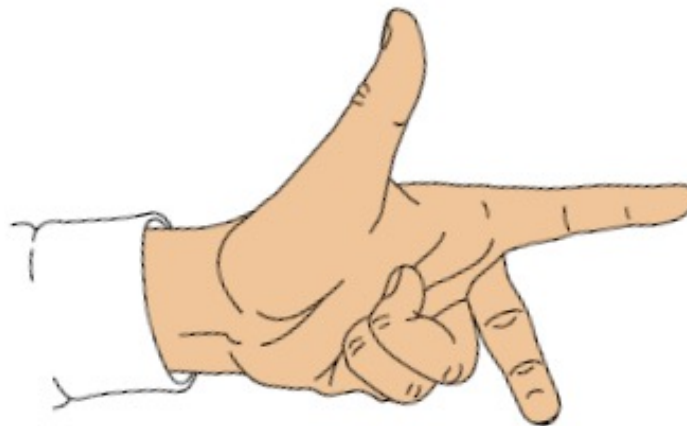


[1] - Gary Bertoline, Eric Wiebe et al, "Technical Graphics Communication", McGraw-Hill Science/Engineering/Math, 4 edition, January 31, 2008, p 312.

Take-Home Message

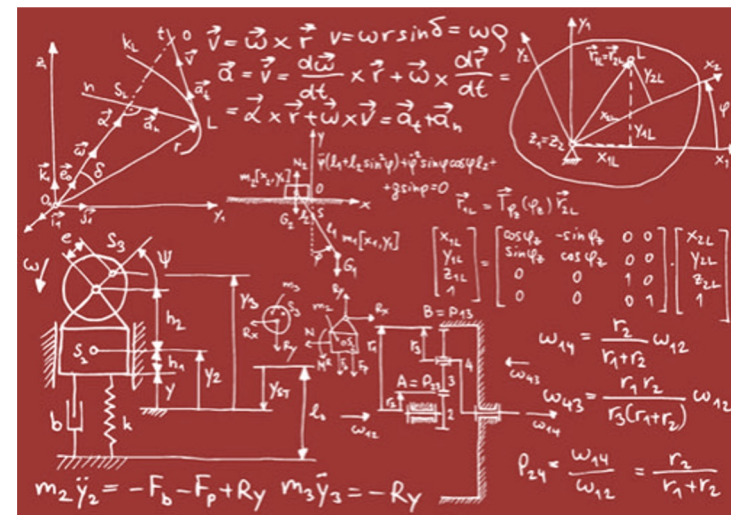
CONCEPT

Right Hand Rule



Vector Mechanics

- ❑ SCALAR & VECTOR
- ❑ VECTOR ALGEBRA
- ❑ VECTOR TYPES
- ❑ TAKE-HOME MESSAGE





Scalar & Vector

SCALAR

What is a scalar?

A scalar is any positive or negative physical quantity that can be completely specified by its magnitude.

Examples?

length, mass, time

How do we represent a scalar?

10 meter, 50 Kilograms, 5 seconds,





Scalar & Vector

VECTORS

What is a vector?

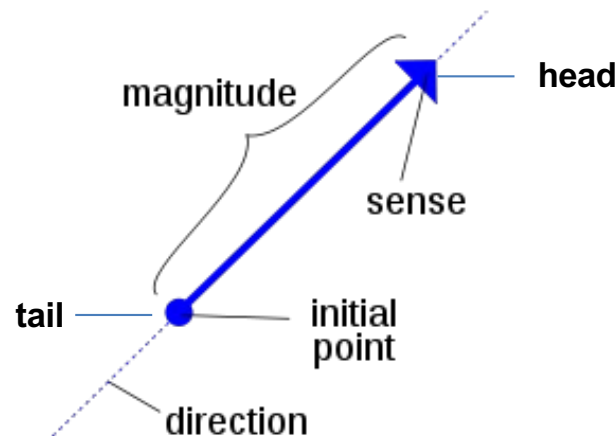
A vector is any physical quantity that requires both a magnitude and a direction for its complete description

Examples?

force, position, velocity, electric current

How do we represent vectors?

Graphically





Vector Algebra

SCALAR & VECTORS

What happens when we multiply a vector with a scalar?

Magnitude increases/decreases scalar times

Negative scalar reverses direction

Can a scalar be added to a vector?

No

How a vector is added to another vector

Triangle rule or “Head to Tail”

Parallelogram law of addition

Cartesian Vector Notation



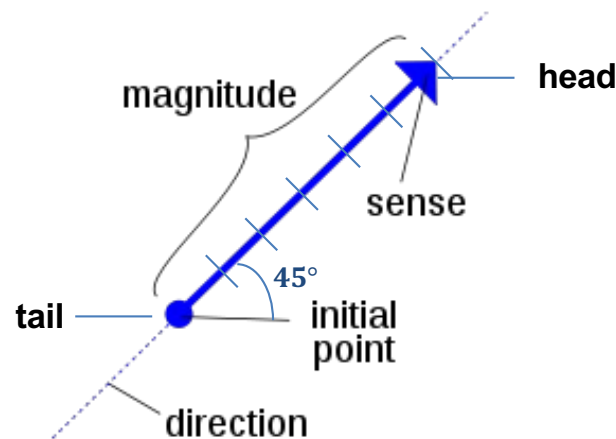
Vector Algebra

MATHEMATICAL FORMULATION

How do we represent vectors mathematically?

Magnitude & Direction

Let's call it
vector $\vec{A}$



$$\text{Magnitude} = |\vec{A}| = 6$$

$$\text{Direction} = \angle \vec{A} = 45^\circ$$

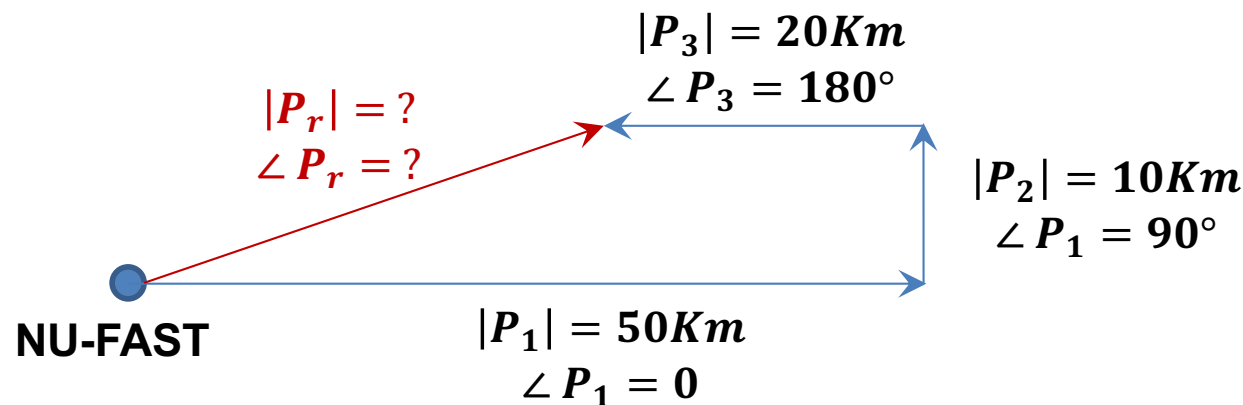


Vector Algebra

EXAMPLE I

Suppose a student was at NU-FAST. He drives 50Km East and then he drives 10Km North followed by a 20km transit toward west.

I want to know, how far is he from campus?



Can I solve it?

Yes! Using Head & Tail Rule + Trigonometry I can?

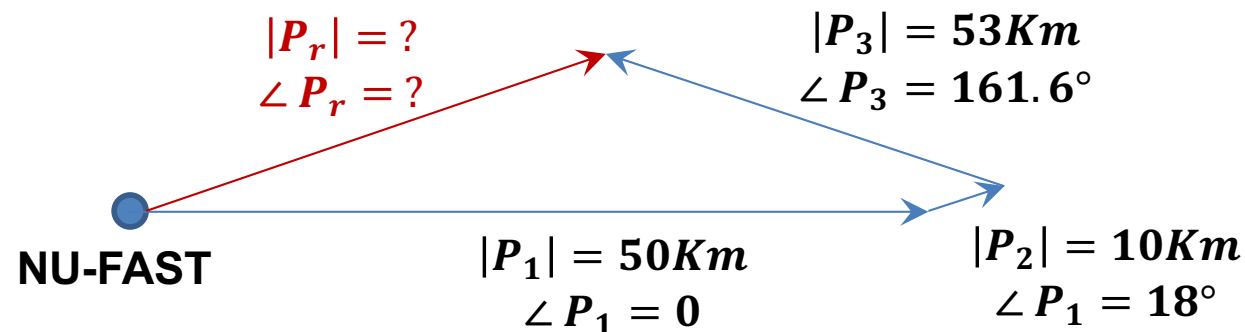
$$|\vec{P}_r| = \sqrt{30^2 + 10^2} = 31.623 Km \quad \angle \vec{P}_r = \tan^{-1} \left(\frac{10}{30} \right) = 18.635^\circ$$



Vector Algebra

EXAMPLE I

What if we were solving the following scenario?



NOT so interesting for analytical work :(

What is the main difficulty?

Arbitrary directions

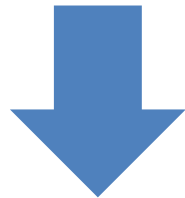
How can we solve this difficulty?



Vector Algebra

SOLUTION

Vectors Addition/Subtraction is similar to Scalar Addition/Subtraction, if they are in the same direction



Resolve vectors on to standard directions

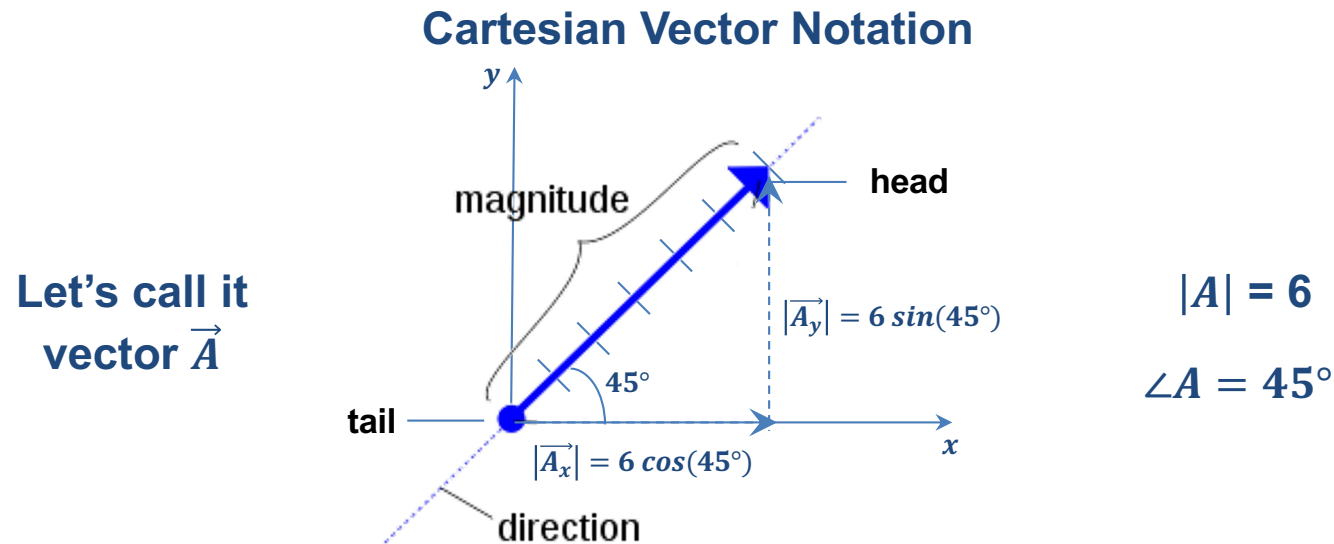




Vector Algebra

CARTESIAN VECTOR NOTATION

How else do we represent vectors mathematically?



$$\vec{A} = |A_x| \hat{i} + |A_y| \hat{j} + |A_z| \hat{k}$$

$$\vec{A} = 6 \cos(45^\circ) \hat{i} + 6 \sin(45^\circ) \hat{j} + 0 \hat{k}$$

$\hat{i}, \hat{j}$ and $\hat{k}$ are unit vectors in the direction of x, y and z-axis respectively

$$\vec{A} = \begin{bmatrix} |A_x| \\ |A_y| \\ |A_z| \end{bmatrix} \quad \vec{A} = \begin{bmatrix} 6 \cos(45^\circ) \\ 6 \sin(45^\circ) \\ 0 \end{bmatrix} = \begin{bmatrix} 4.243 \\ 4.243 \\ 0 \end{bmatrix}$$

Preferred writing style!

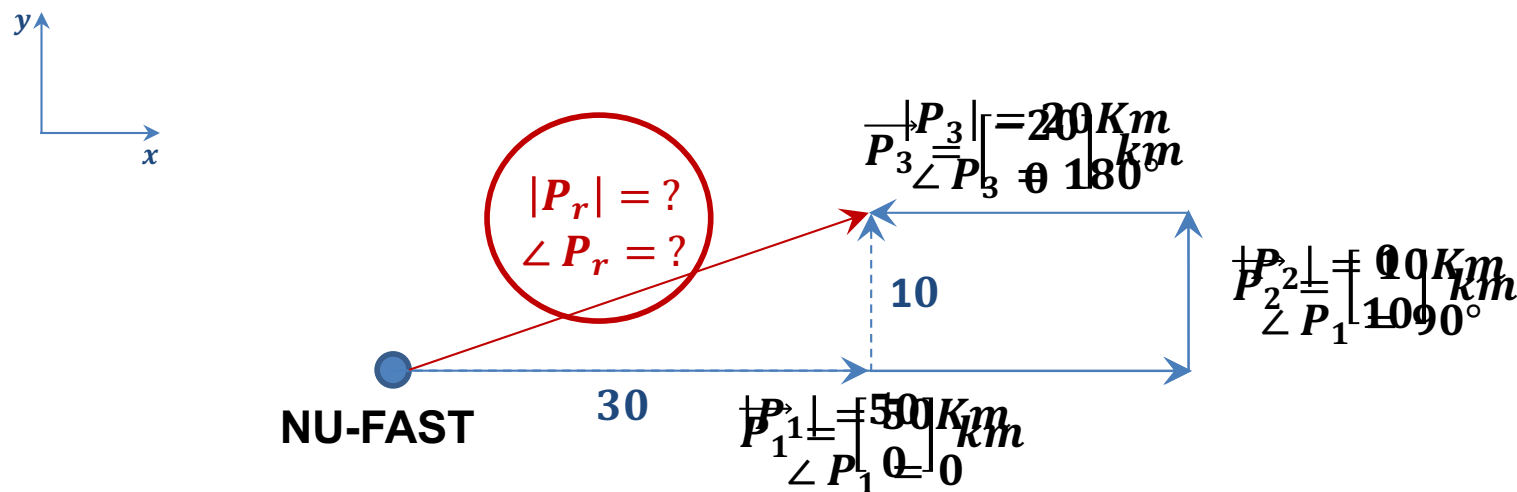


Vector Algebra

EXAMPLE I CONT.

Suppose a student was at NU-FAST. He drives 50Km East and then he drives 10Km North followed by a 20km transit toward west.

I want to know, how far is he from campus?



$$\vec{P}_r = \vec{P}_1 + \vec{P}_2 + \vec{P}_3$$

$$\vec{P}_r = \begin{bmatrix} 50 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 10 \end{bmatrix} + \begin{bmatrix} -20 \\ 0 \end{bmatrix} = \begin{bmatrix} 30 \\ 10 \end{bmatrix} \text{ km}$$



Vector Algebra

MAGNITUDE & DIRECTION

Given rectangular components, how can we find the actual vector?

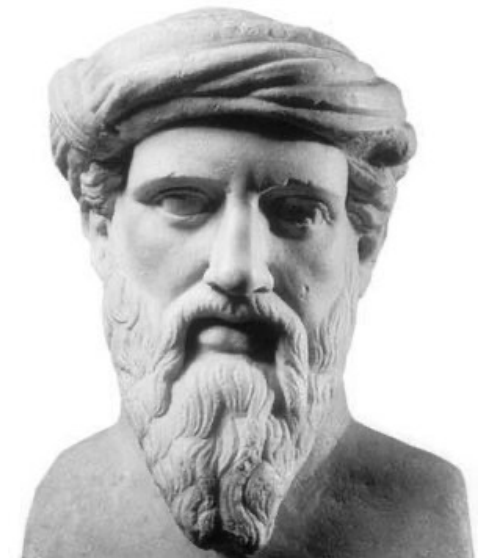
Pythagorean theorem

$$|\vec{A}| = \sqrt{|\vec{A}_x|^2 + |\vec{A}_y|^2}$$

$$\angle \vec{A} = \tan^{-1} \left(\frac{|\vec{A}_y|}{|\vec{A}_x|} \right)$$

$$|\vec{A}| = \sqrt{|\vec{A}_x|^2 + |\vec{A}_y|^2 + |\vec{A}_z|^2}$$

$$\cos \alpha = \frac{|\vec{A}_x|}{|\vec{A}|} \quad \cos \beta = \frac{|\vec{A}_y|}{|\vec{A}|} \quad \cos \gamma = \frac{|\vec{A}_z|}{|\vec{A}|}$$



**Pythagoras of Samos
(570 -495 BC)**

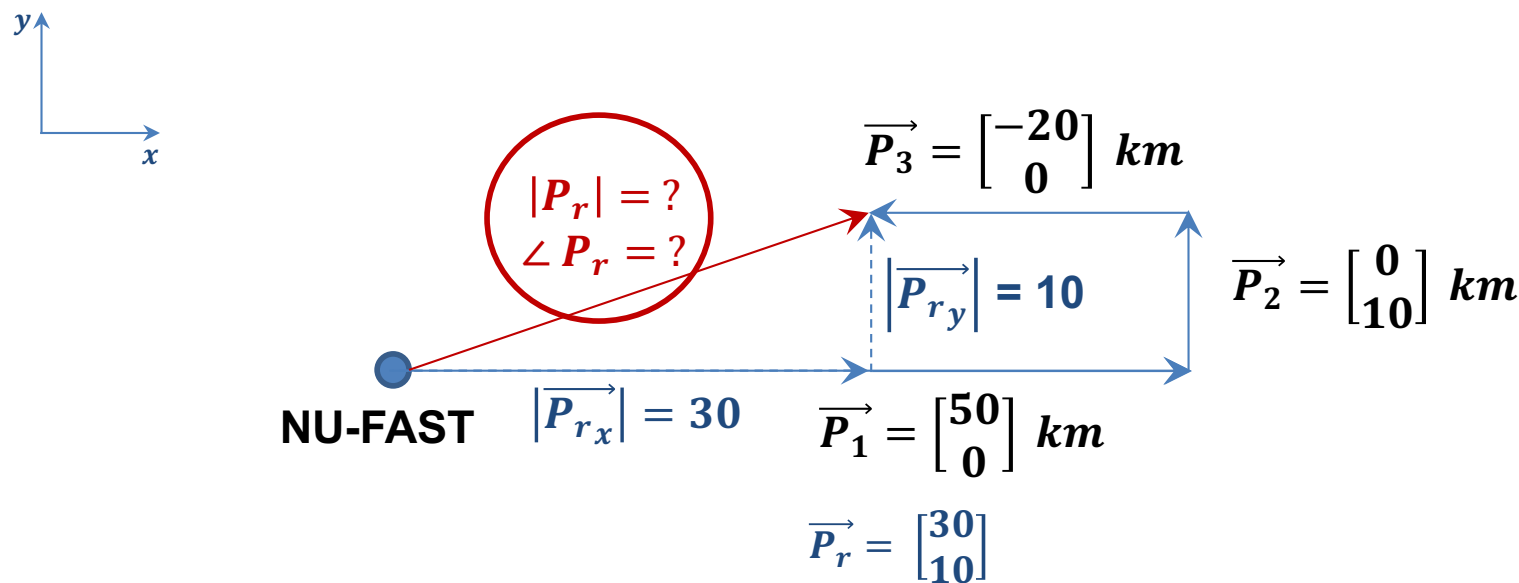


Vector Algebra

EXAMPLE I CONT.

Suppose a student was at NU-FAST. He drives 50Km East and then he drives 10Km North followed by a 20km transit toward west.

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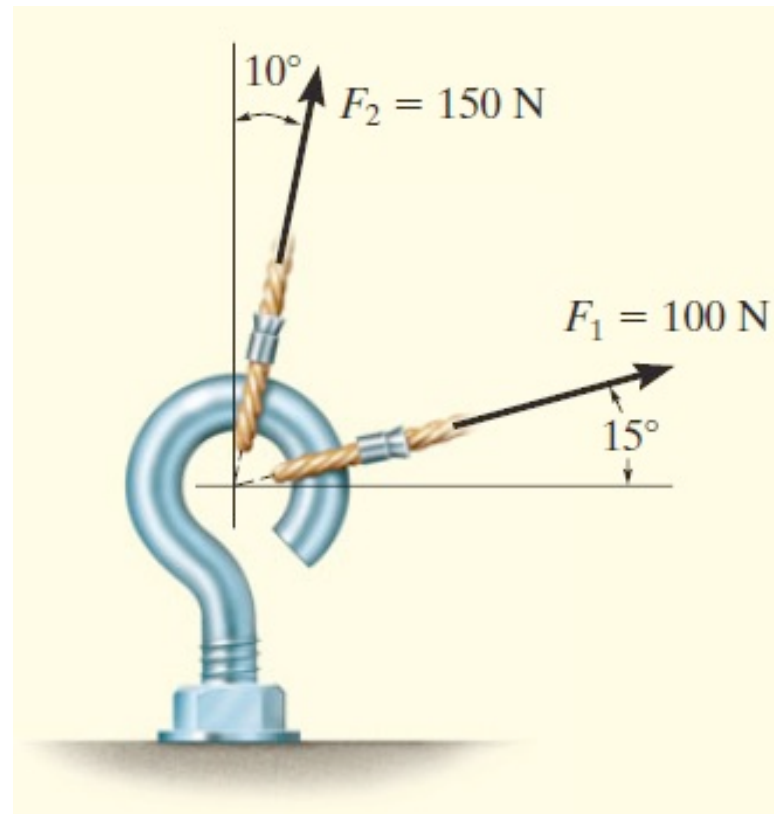
$$|\vec{P}_r| = \sqrt{|\vec{P}_{rx}|^2 + |\vec{P}_{ry}|^2} = \sqrt{30^2 + 10^2} = 31.623 \text{ Km}$$

$$\angle \vec{P}_r = \tan^{-1} \left(\frac{|\vec{P}_{ry}|}{|\vec{P}_{rx}|} \right) = \tan^{-1} \left(\frac{10}{30} \right) = 18.635^\circ$$



Vector Algebra

EXAMPLE II



[1]

Determine the magnitude and direction of the resultant force.

Parallelogram Law

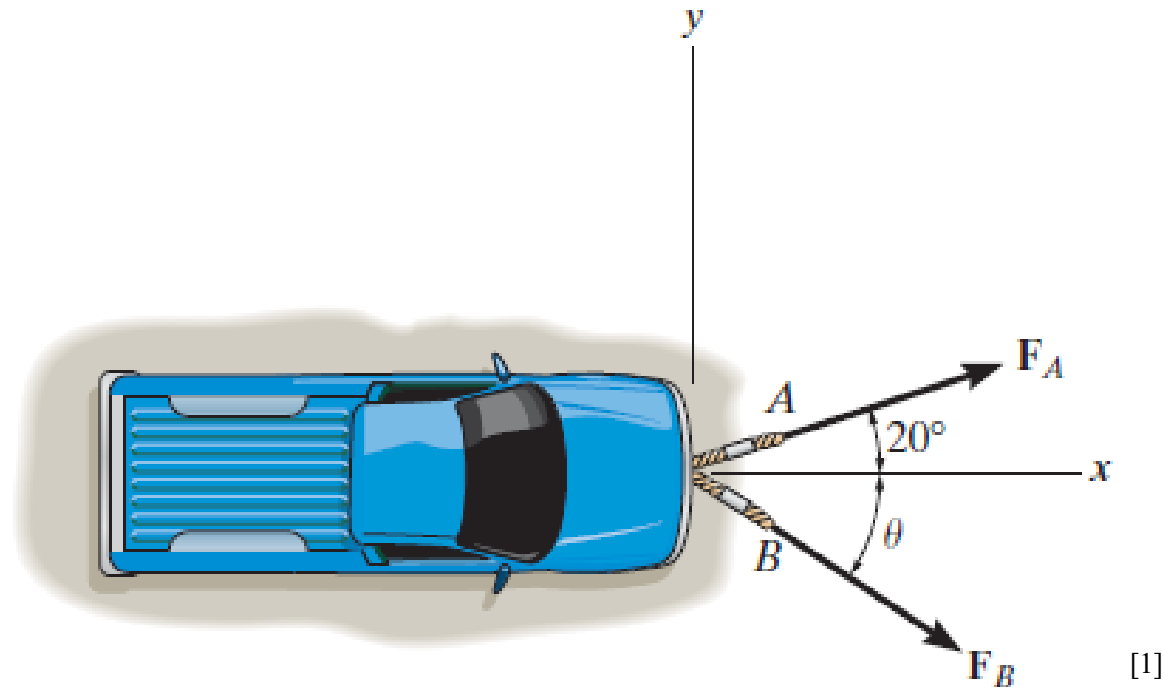
Cartesian Vector Notation

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, Example 2.1



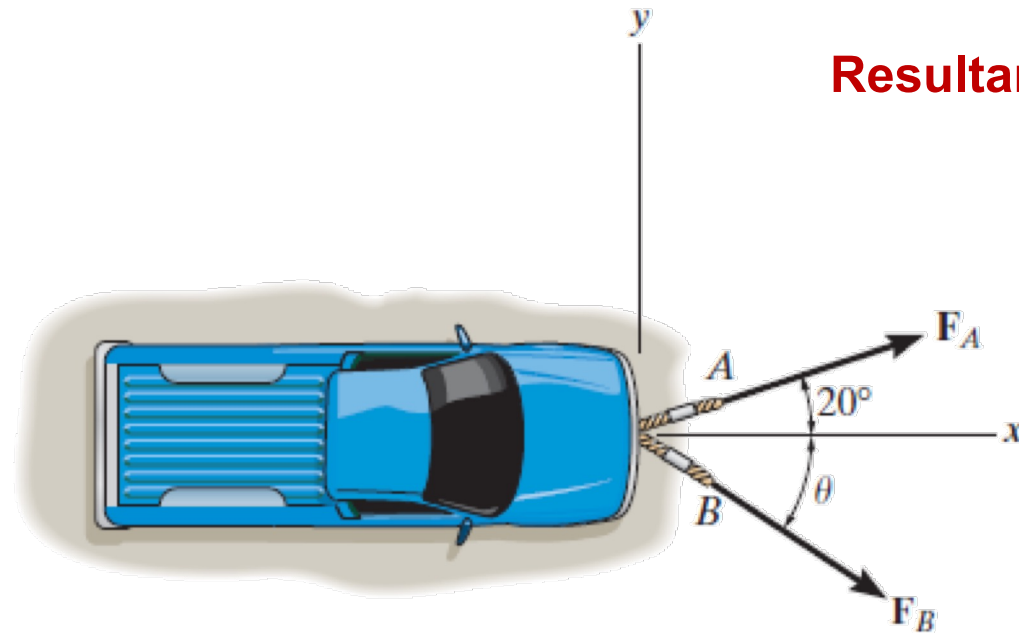
Vector Algebra

EXAMPLE III



The truck is to be towed using two ropes. Determine the magnitudes of forces F_A and F_B acting on each rope in order to develop a resultant force of 950 N directed along the positive x-axis. Set $\theta = 50^\circ$.

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, Problem 2.18



Resultant force of 950 N

$\theta = 50^\circ$.



Vector Algebra

UNIT VECTOR

What is a Unit Vector?

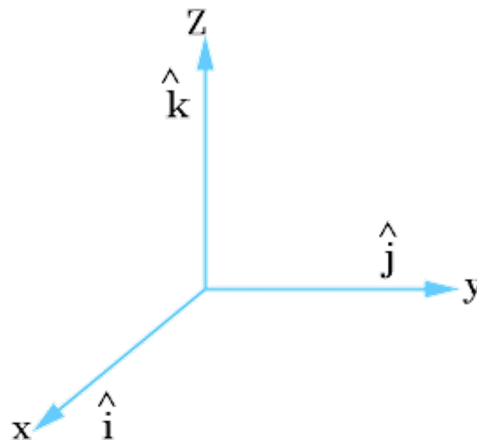
A vector of Unit Magnitude

How to find a Unit Vector along a given vector $\vec{A}$?

Dividing the given vector by its magnitude

$$\hat{A} = \frac{\vec{A}}{|\vec{A}|}$$

A unit vector is often used to represent directions

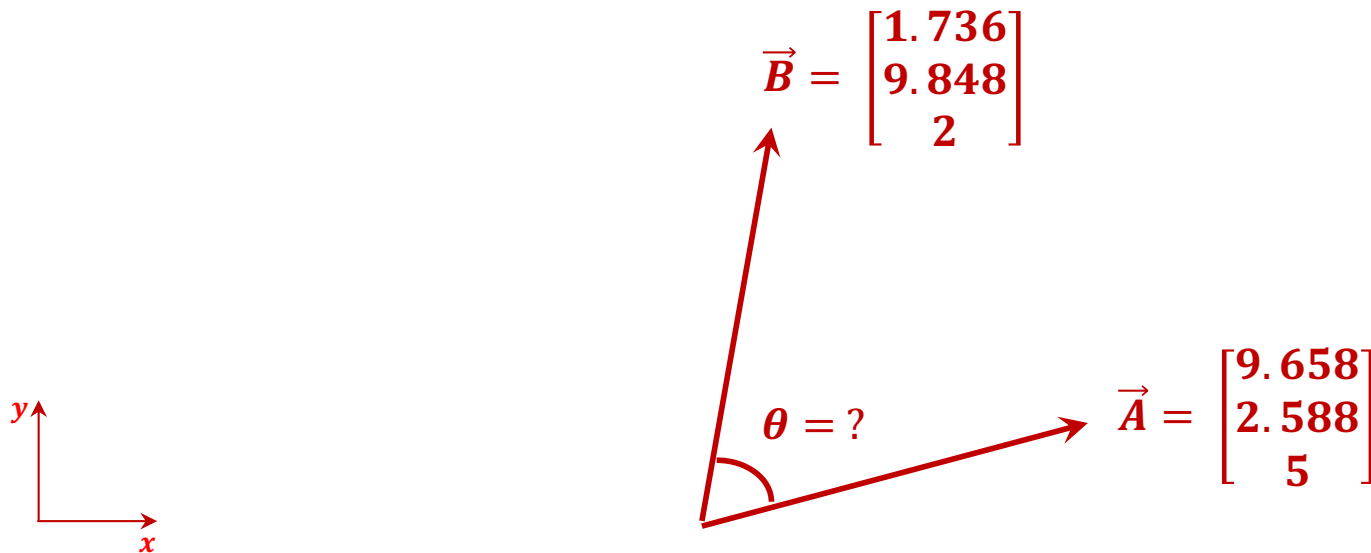




Vector Algebra

VECTOR PRODUCTS

Given two vectors, how can one find the angle between them?



$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos(\theta)$$

$$\vec{A} \cdot \vec{B} = 52.253$$

$$|\vec{A}| = 10.198$$

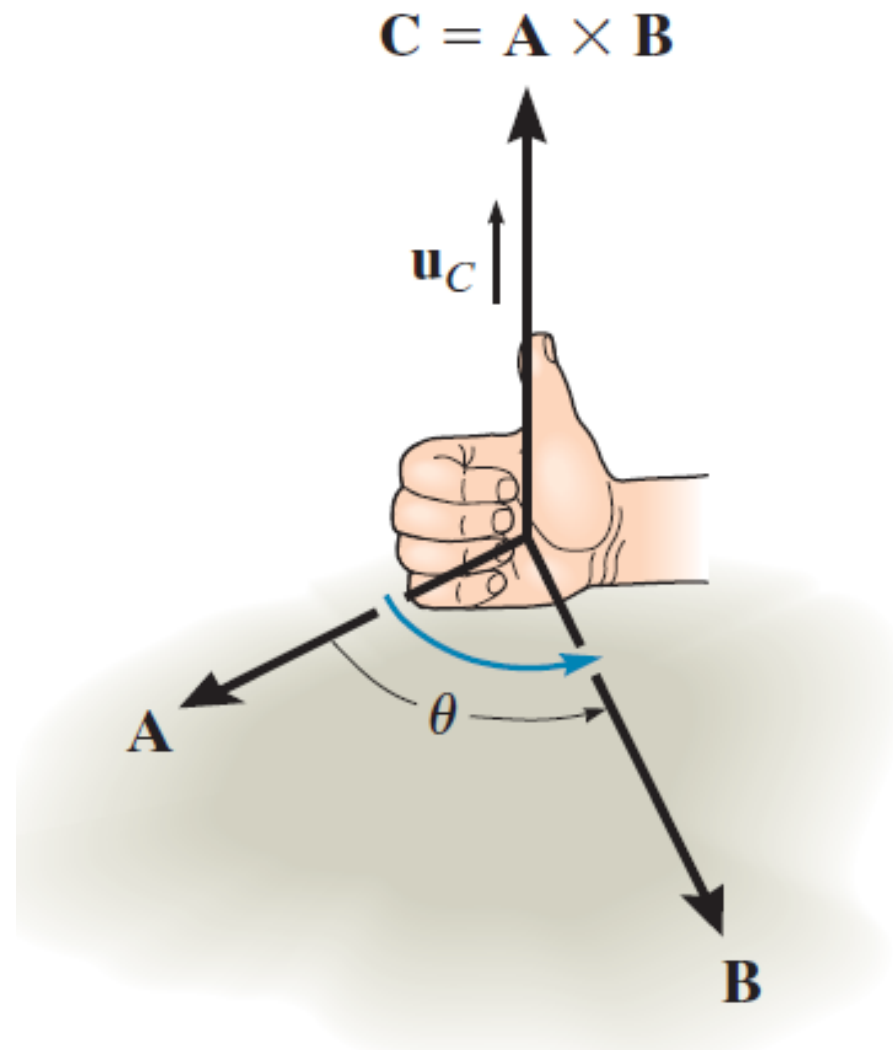
$$|\vec{B}| = 11.179$$

$$\theta = 62.72^\circ$$



Vector Algebra

VECTOR PRODUCTS

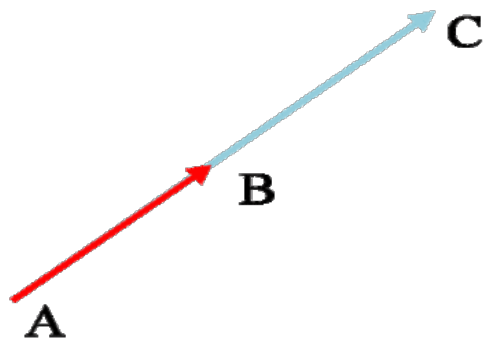




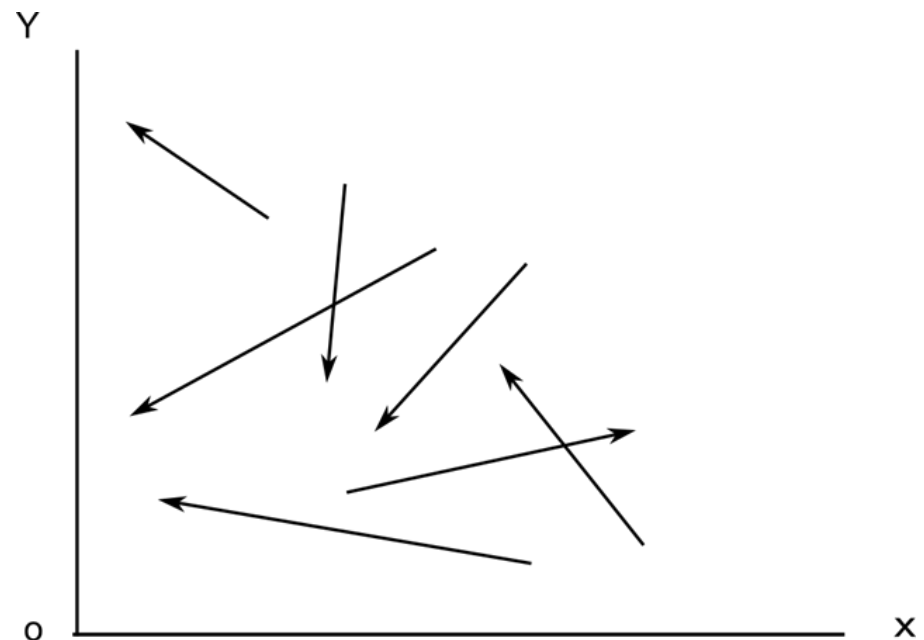
Vector Types

COLLINEAR

Collinear vectors are along the same line



Collinear



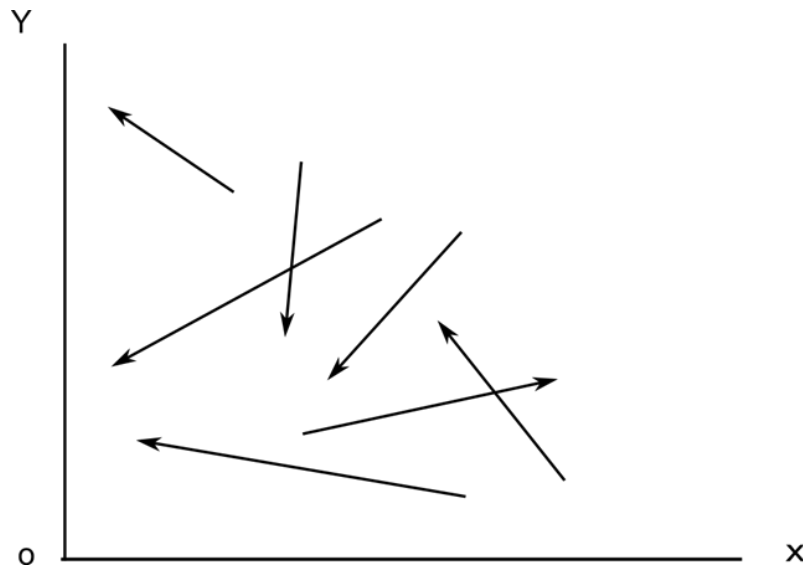
NOT Collinear



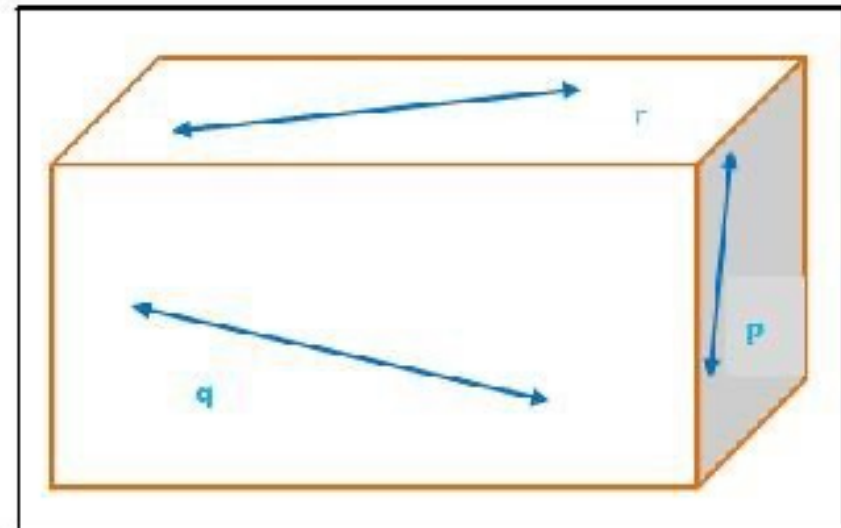
Vector Types

COPLANAR

Coplanar vectors in the same plane



Coplanar



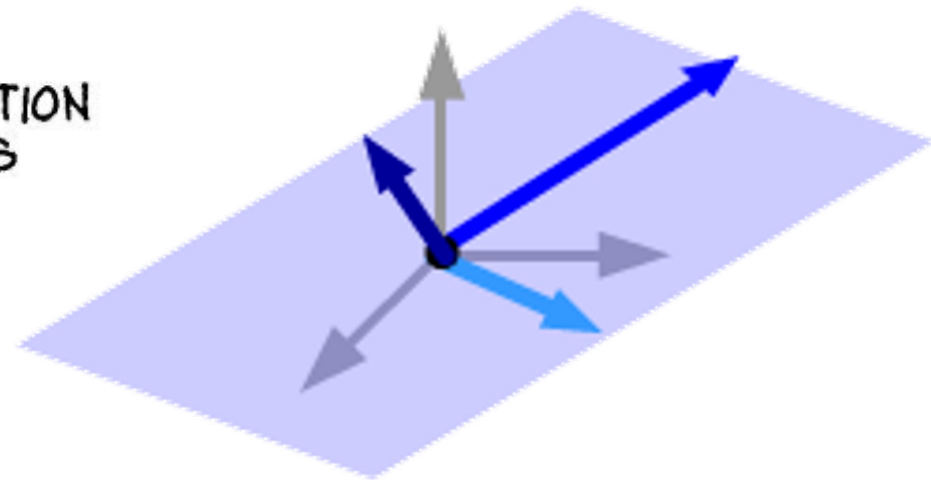
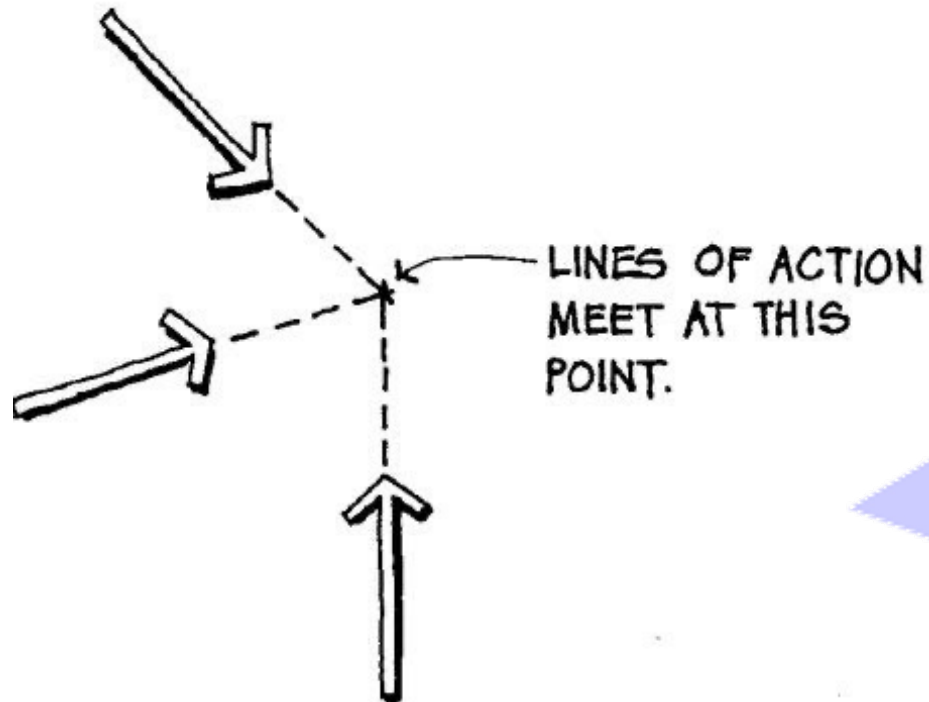
NOT Coplanar



Vector Types

CONCURRENT

Concurrent vector system is one in which the lines of action of all the vectors intersect at a common point O





Take-Home Message

PROBLEM SOLVING STEPS

1 – Mathematical Formulation

2 – Solution Strategy

3 – Actual Solution

4 – Justification & Reflection



Take-Home Message

CARTESIAN VECTOR NOTATION

**Vectors Addition/Subtraction is similar to
Scalar Addition/Subtraction, if they are in
the same direction**



Take-Home Message

APPLICATION OF DOT PRODUCT

**1 - Finding the angle between two
given vectors**

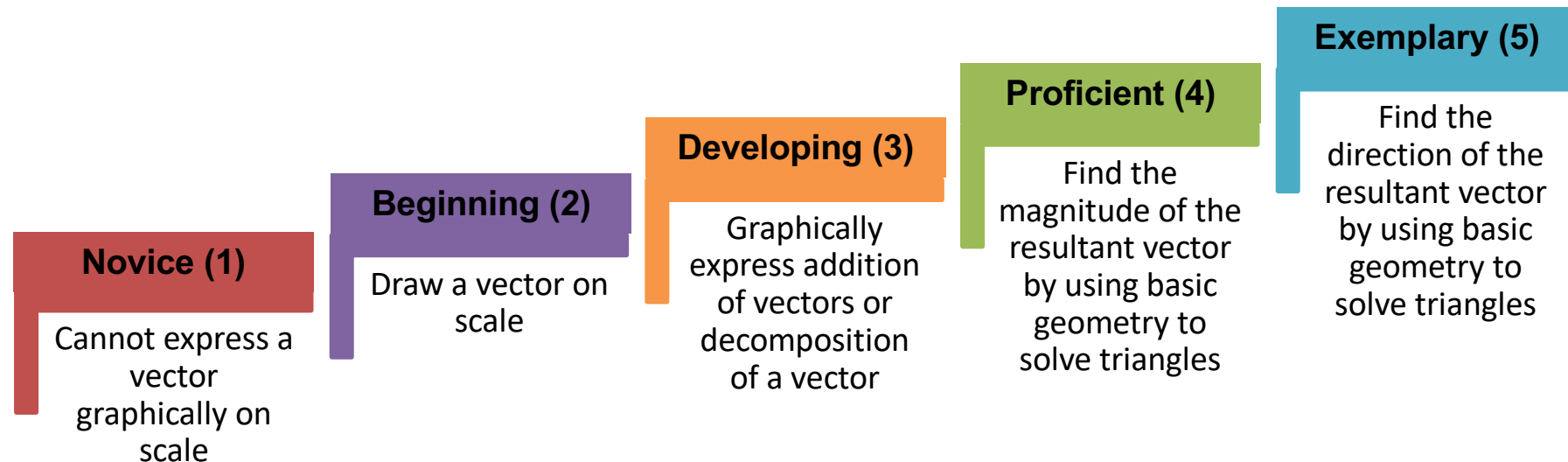
**2 – Finding the projection of one
vector on to another**



Course Learning Outcome

CLO1

Student should be able to Graphically add or decompose vector quantities

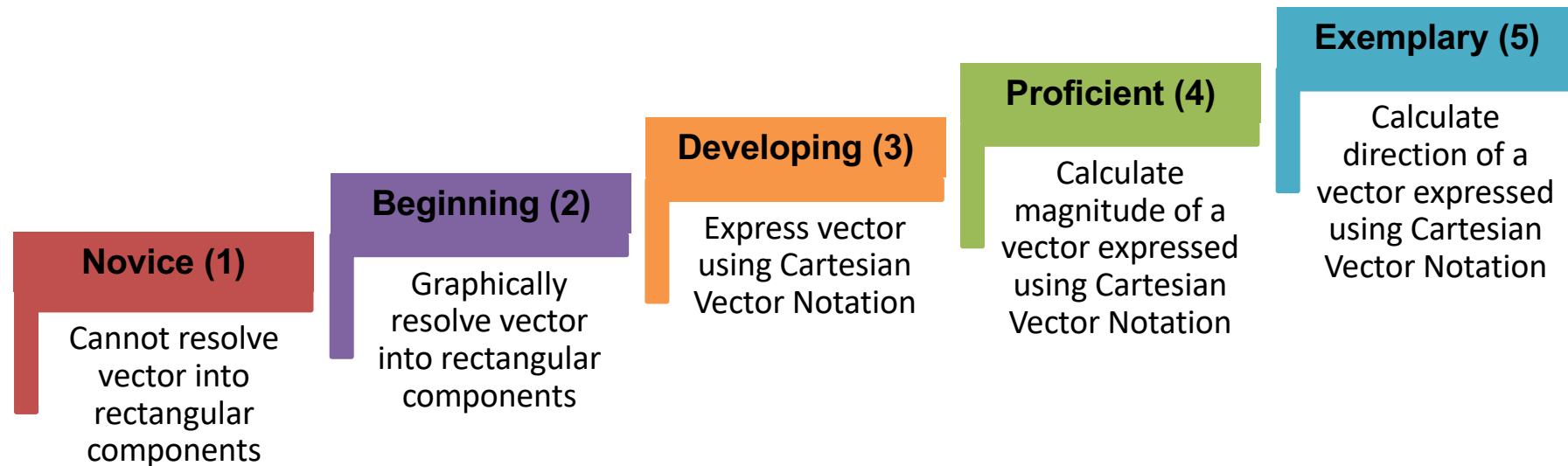




Course Learning Outcome

CLO2

Resolve a vector into rectangular components, express it in Cartesian Vector Notation and calculate its magnitude and direction.

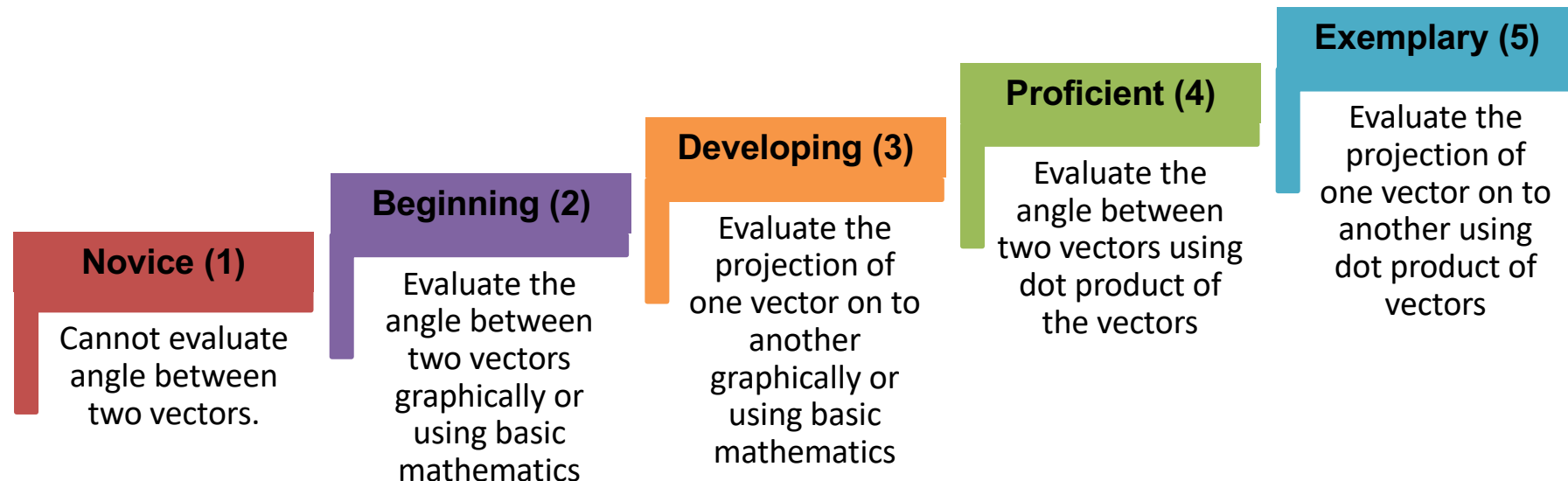




Course Learning Outcome

CLO3

Evaluate the angle between two vectors and the projection of one onto the other.



Force Systems

- ☐ FORCE SYSTEMS
- ☐ MOMENTS OR TORQUE
- ☐ COUPLE MOMENT





Force Systems

NEWTON'S LAWS OF MOTION

What is Newton's First Law of Motion?

A body at rest remains at rest and a body in motion continues its motion in a straight line unless acted upon by an external force

What is Newton's Second Law of Motion?

The force action on a body gives it an acceleration which is in the direction of force

$$F = m a$$

What is Newton's Third Law of Motion?

If one object exerts a force on another object, the other exerts the same force in the opposite direction on the one

Action = - Reaction

Contact force



Force Systems

MOMENTS OR TORQUE

What is the tendency of a force to produce rotary motion in a body called?

Moment or Torque

What are the factors on which the magnitude of moment depends?

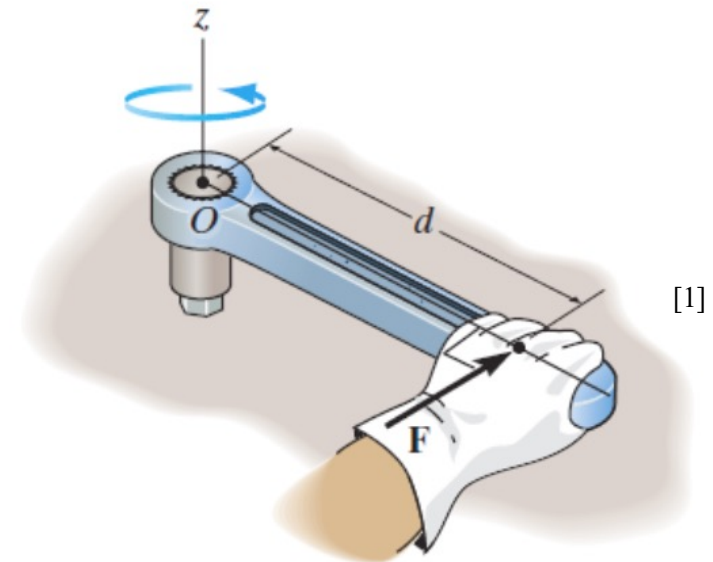
Magnitude of force, F

Perpendicular distance from axis of rotation d

$$M = F d$$

The magnitude of the moment is directly proportional to the magnitude of Force F and the perpendicular distance or moment arm d

Moment is defined with respect to a fix reference point



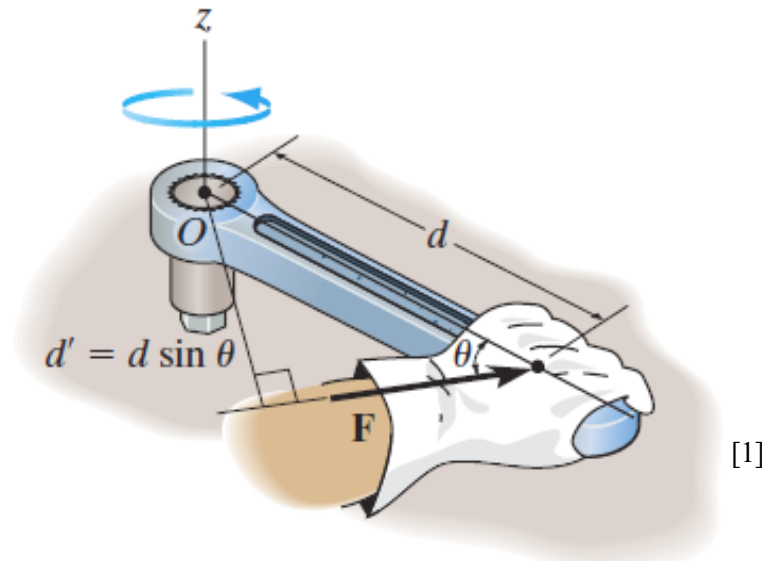
[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, pp. 117



Moments or Torque

BASIC CONCEPTS

What if the force is applied at an angle?



The component of distance is taken, which is perpendicular to the force

Or the component of force is taken, which is perpendicular to the distance

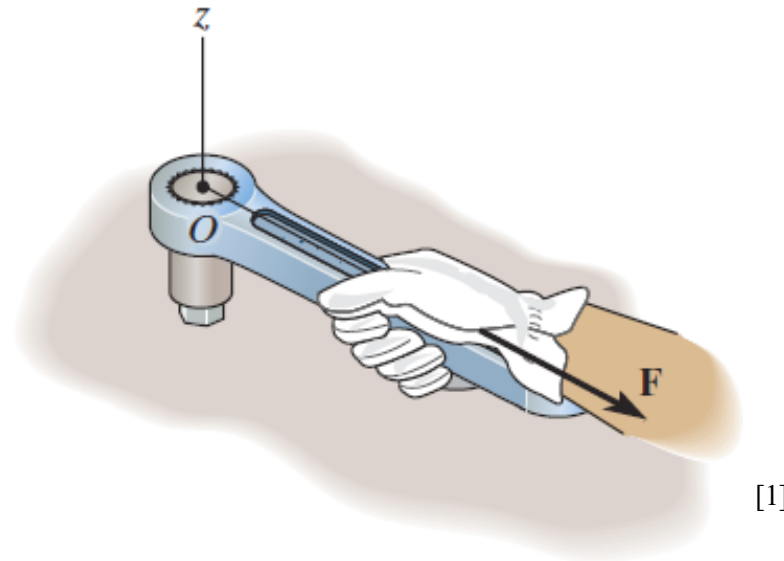
[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, pp. 117



Moments or Torque

BASIC CONCEPTS

What if the force is passing through the reference point?



No Moment

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, pp. 117



Moments or Torque

VECTOR NOTATION

Is moment a scalar or a vector?

Vector

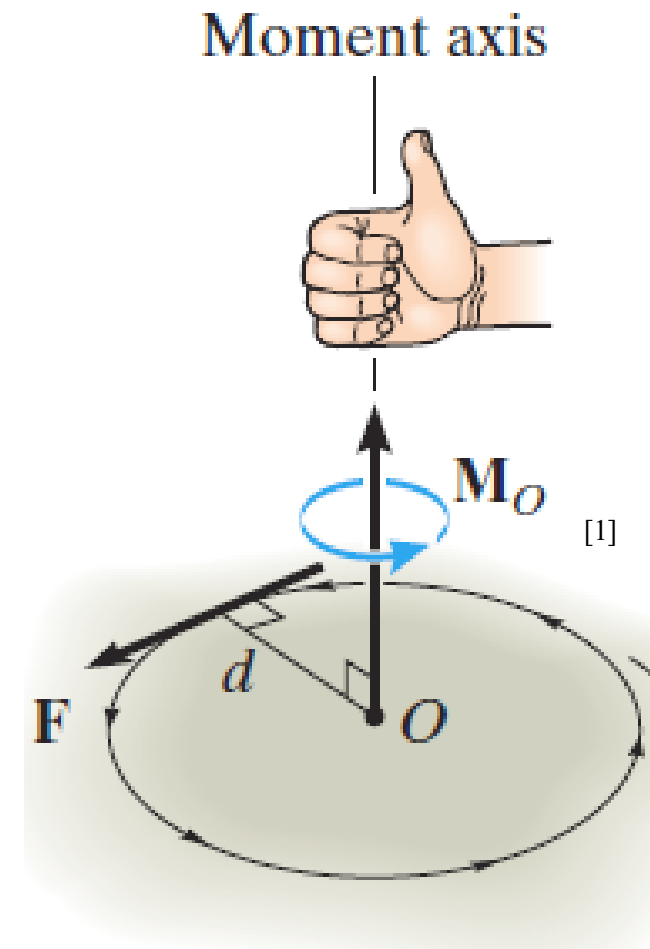
What about its direction?

It is perpendicular to the plane that contains the force and its moment arm

Which mathematical operator, when applied to two vectors, give another vector which is perpendicular to both?

$$\underline{M}_O = \underline{r} \times \underline{F}$$

r can be any position vector, measured from point O to any point on the line of action of the force F



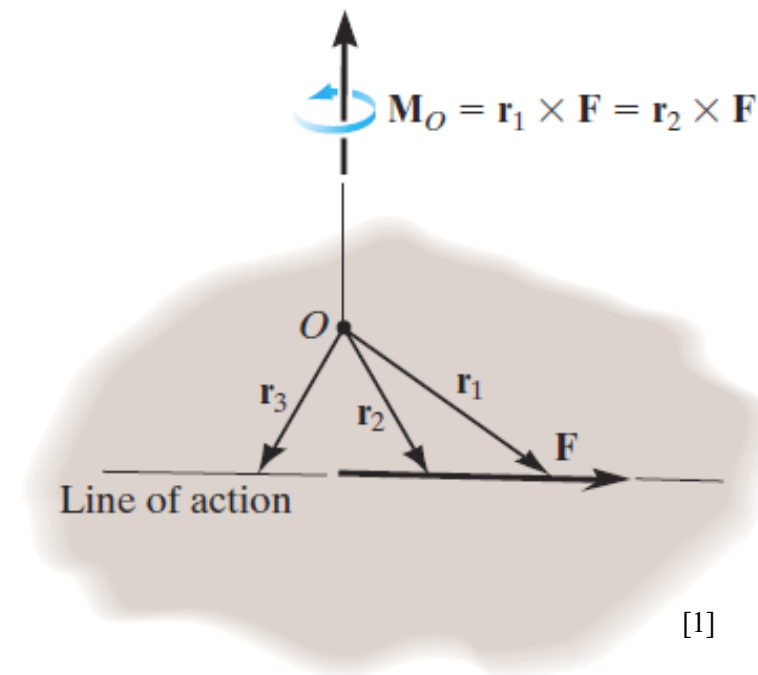
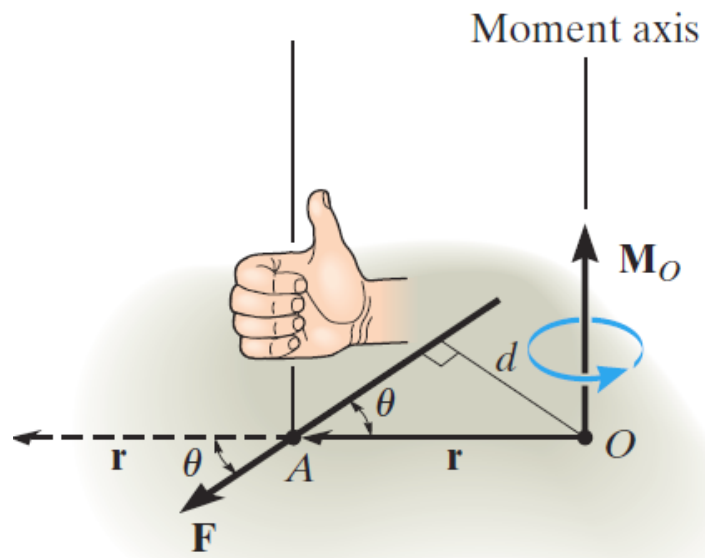
[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, pp. 118



Moments or Torque

PRINCIPLE OF TRANSMISSIBILITY

$$\underline{M}_O = \underline{r} \times \underline{F}$$



**F can be applied at any point along its line of action
and still create this same moment about point O**

This property is called the principle of transmissibility of a force

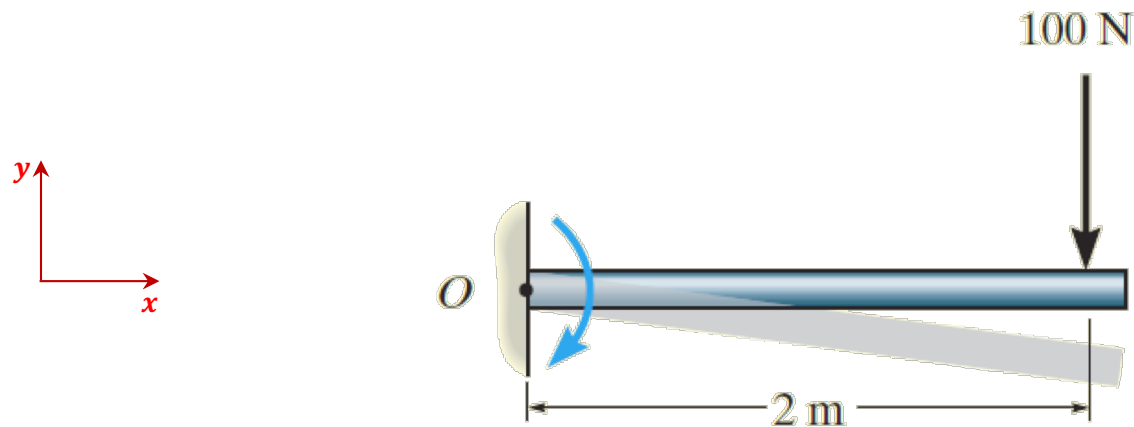
[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, pp. 118



Moments or Torque

EXAMPLE I

Calculate moment about point O



Magnitude:

$$M_o = F d = (100)(2) = 200 \text{ Nm}$$

Direction:

-ve z-axis

$$\underline{r} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \text{ m} \quad \underline{F} = \begin{bmatrix} 0 \\ -100 \\ 0 \end{bmatrix} \text{ N}$$

$$\underline{M}_o = \underline{r} \times \underline{F}$$

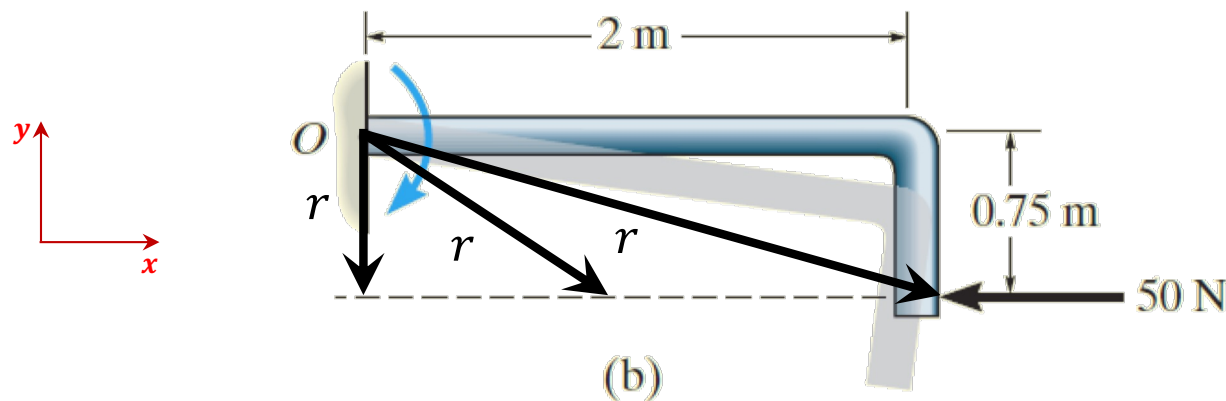
$$\underline{M}_o = \begin{bmatrix} 0 \\ 0 \\ -200 \end{bmatrix} \text{ Nm}$$



Moments or Torque

EXAMPLE II

Calculate moment about point O



Magnitude:

$$M_o = F d = (50)(0.75) = 37.5 \text{ Nm}$$

Direction:

-ve z-axis

$$\underline{r} = \begin{bmatrix} 2 \\ -0.75 \\ 0 \end{bmatrix} \text{ m} \quad \underline{F} = \begin{bmatrix} -50 \\ 0 \\ 0 \end{bmatrix} \text{ N}$$

$$\underline{M}_o = \underline{r} \times \underline{F}$$

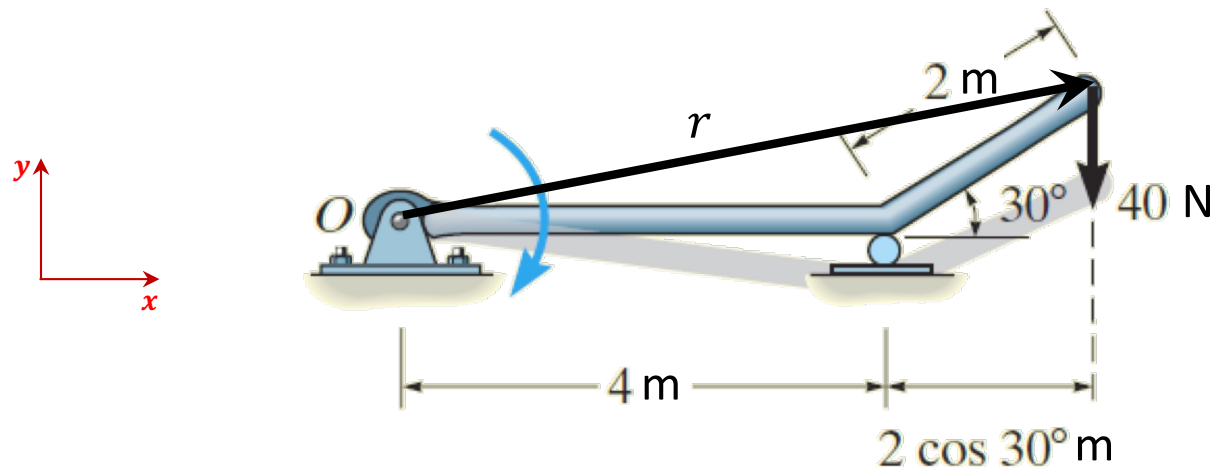
$$\underline{M}_o = \begin{bmatrix} 0 \\ 0 \\ -37.5 \end{bmatrix} \text{ Nm}$$



Moments or Torque

EXAMPLE III

Calculate moment about point O



Magnitude:

$$M_o = F d = (40)(4 + 2\cos(30^\circ)) = 229.282 \text{ Nm}$$

Direction:

-ve z-axis

$$\underline{r} = \begin{bmatrix} 5.732 \\ 1 \\ 0 \end{bmatrix} \text{ m} \quad \underline{F} = \begin{bmatrix} 0 \\ -40 \\ 0 \end{bmatrix} \text{ N}$$

$$\underline{M}_o = \underline{r} \times \underline{F}$$

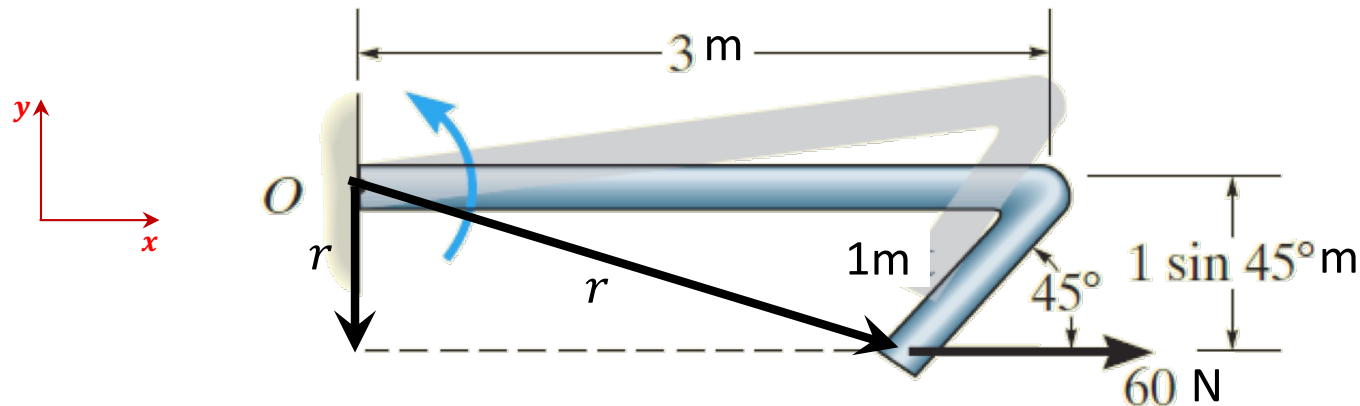
$$\underline{M}_o = \begin{bmatrix} 0 \\ 0 \\ -229.283 \end{bmatrix} \text{ Nm}$$



Moments or Torque

EXAMPLE IV

Calculate moment about point O



Magnitude:

$$M_o = F d = (60)(1 \sin(45^\circ)) = 42.426 \text{ Nm}$$

Direction:

+ve z-axis

$$\underline{r} = \begin{bmatrix} 3 - \cos 45^\circ \\ -\sin 45^\circ \\ 0 \end{bmatrix} \text{ m} \quad \underline{F} = \begin{bmatrix} 60 \\ 0 \\ 0 \end{bmatrix} \text{ N}$$

$$\underline{M}_o = \underline{r} \times \underline{F}$$

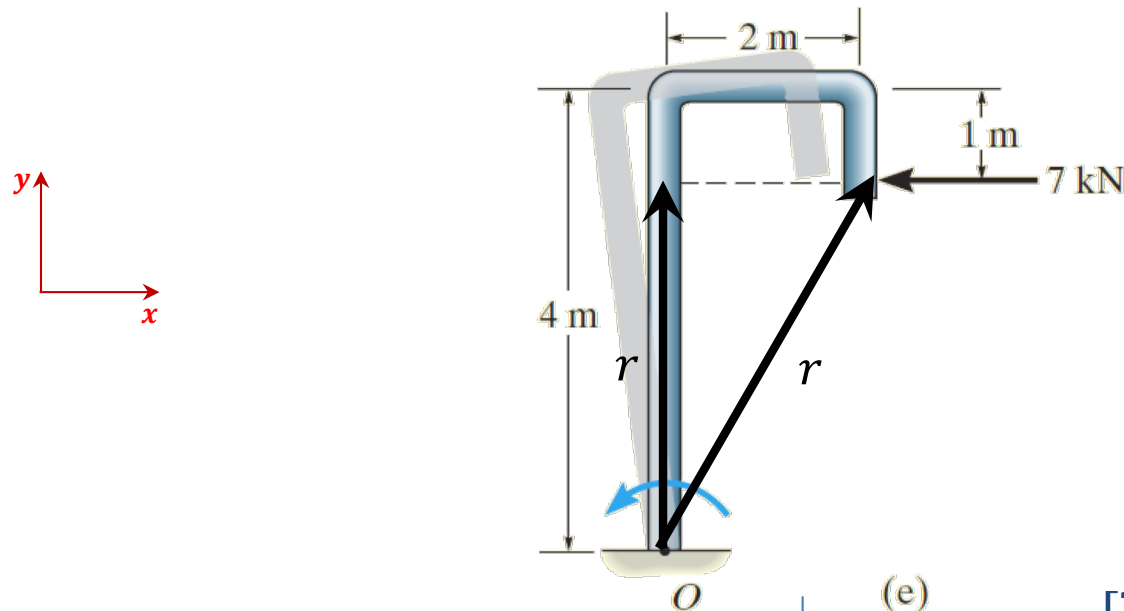
$$\underline{M}_o = \begin{bmatrix} 0 \\ 0 \\ 42.426 \end{bmatrix} \text{ Nm}$$



Moments or Torque

EXAMPLE V

Calculate moment about point O



Magnitude:

$$M_o = F d = (7000)(4 - 1) = 21 \text{ kNm}$$

Direction:

+ z-axis

(e)

$$\underline{r} = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} \text{ m} \quad \underline{F} = \begin{bmatrix} -7000 \\ 0 \\ 0 \end{bmatrix} \text{ N}$$

$$\underline{M}_o = \underline{r} \times \underline{F}$$

$$\underline{M}_o = \begin{bmatrix} 0 \\ 0 \\ 21000 \end{bmatrix} \text{ Nm}$$